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Green leaf volatile production by plants: a meta-analysis

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Summary

Plants respond to stress by releasing biogenic volatile organic compounds (BVOCs). Green leaf volatiles (GLVs), which are abundantly produced across the plant kingdom, comprise an important group within the BVOCs. They can repel or attract herbivores and their natural enemies; and they can induce plant defences or prime plants for enhanced defence against herbivores and pathogens and can have direct toxic effects on bacteria and fungi. Unlike other volatiles, GLVs are released almost instantly upon mechanical damage and (a)biotic stress and could thus function as an immediate and informative signal for many organisms in the plant's environment. We used a meta-analysis approach in which data from the literature on GLV production during biotic stress responses were compiled and interpreted. We identified that different types of attackers and feeding styles add a degree of complexity to the amount of emitted GLVs, compared with wounding alone. This meta-analysis illustrates that there is less variation in the GLV profile than we presumed, that pathogens induce more GLVs than insects and wounding, and that there are clear differences in GLV emission between monocots and dicots. Besides the meta-analysis, this review provides an update on recent insights into the perception and signalling of GLVs in plants.

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See also the Editorial by Kessler, **220**: 655–658.

I. Introduction

Green leaf volatiles (GLVs) represent an important group of plant volatiles. They consist of six carbon (C6) compounds including alcohols, aldehydes and esters (Dudareva *et al.*, 2006) and are released from almost every plant. Curtius & Franzen (1914) were the first to isolate *E*-2-hexenal (*E*-2-HAL) from 600 kg of European hornbeam leaves. Other GLVs have been identified and isolated thereafter (Hatanaka *et al.*, 1987). The release of GLVs is caused by mechanical damage (Halitschke *et al.*, 2004) or herbivory (Allmann *et al.*, 2013), by fungal or bacterial infection (Piesik *et al.*, 2011; Ponzio *et al.*, 2013) but also as a consequence of abiotic stress such as drought (Wenda-Piesik, 2011), heat (Copolovici *et al.*, 2012), high light (Loreto *et al.*, 2006) and presence of heavy metals (Obara *et al.*, 2002) (Fig. 1). This general response after (a)biotic stress suggests an important role for GLVs in plants in reaction to stressful environments. GLVs are implicated in a panoply of interactions; they have been reported to repel or attract herbivores and their natural enemies (Visser *et al.*, 1979; Turlings *et al.*, 1991; Scala *et al.*, 2013), to activate and prime plant defences (Engelberth *et al.*, 2004; Ameye *et al.*, 2015), to activate abiotic-stress related genes (Yamauchi *et al.*, 2015), and to have antibacterial and antifungal properties (Nakamura & Hatanaka, 2002; Kishimoto *et al.*, 2008) (Fig. 1).

The widespread GLV production as a general response to different types of biotic stress prompts the question as to whether the type of biotic attack moulds the outcome and pattern of GLV production. To answer this question, we performed a meta-analysis on the available literature. Compared with a traditional review, a meta-analysis has the advantage of adding a holistic, generalized, systematic dimension. It offers the benefit of providing a quantitative synthesis on available data or literature and allows researchers to assess between-study variability and identify factors that determine the treatment effects on the physiology of plants (Trikalinos *et al.*, 2008).

The analysis reveals a clear interaction among GLV production, the type of biotic attacker, the type of feeding guild and the class of plants. In addition to the meta-analysis, we give an update on the current status of GLV research regarding biosynthesis, perception and signal transduction, and discuss the biological function of GLVs.

II. Biosynthesis

Green leaf volatiles are generated through the oxylipin pathway from C18-polyunsaturated fatty acids (FAs; α -linolenic acid (ALA) and linoleic acid; Fig. 2) (Matsui, 2006). While it has been assumed that FAs are hydrolysed from membrane lipids (Matsui *et al.*, 2000) before further enzymatic reaction, little is known about the identity

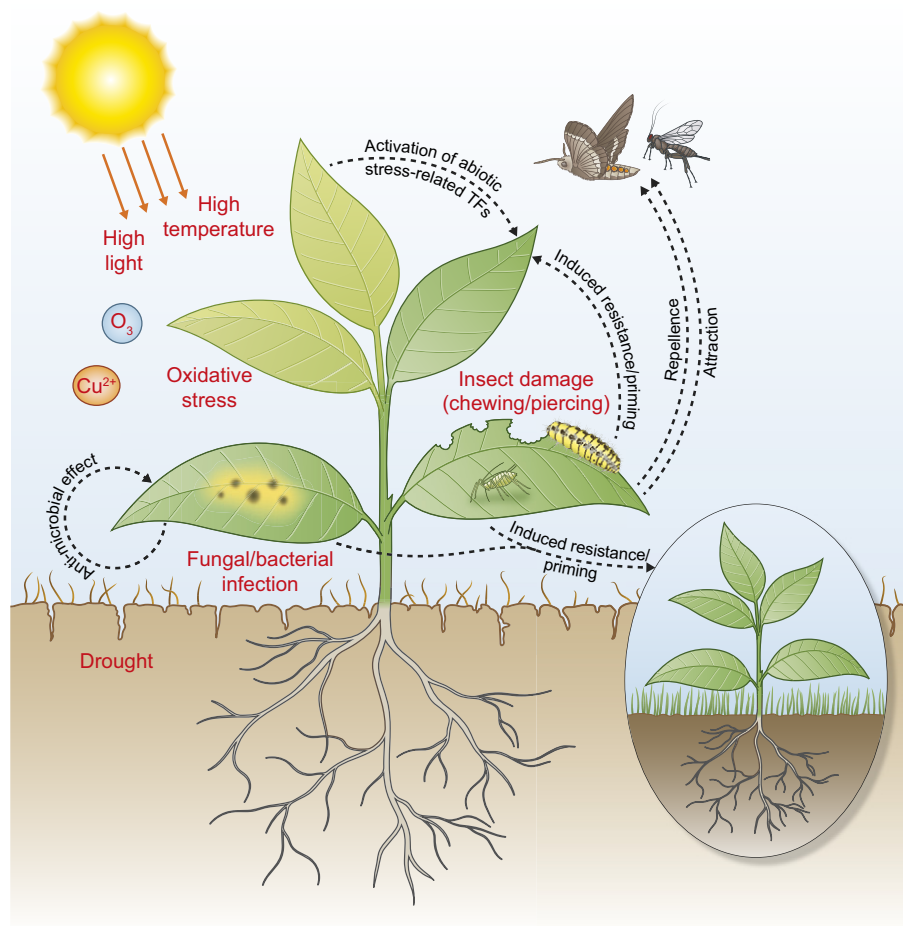


Fig. 1 Functions of green leaf volatiles (GLVs). Overview of the different types of (a)biotic stress that drive GLV production (red) and reported functions of GLVs (black). See the text for references.

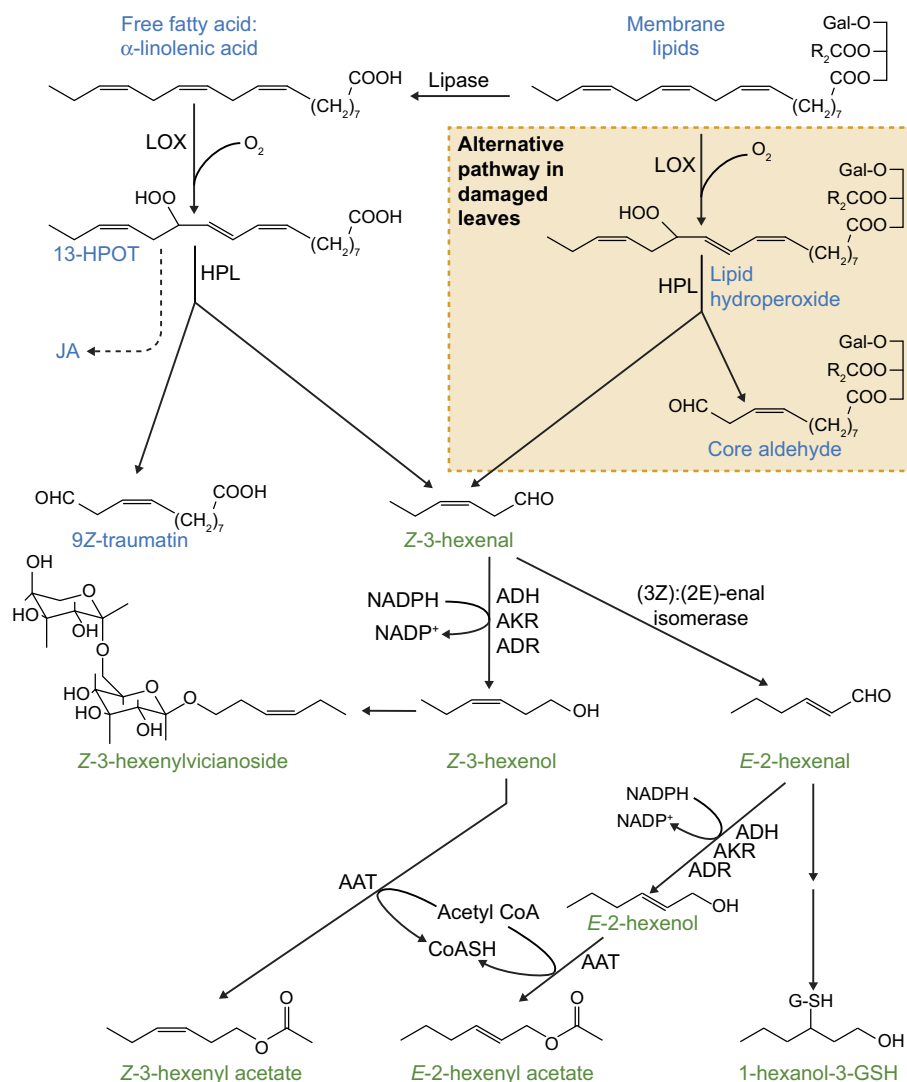


Fig. 2 Biosynthetic pathway of green leaf volatiles (GLVs). Free fatty acids (FAs) are released from membrane lipids. The two FAs that serve as substrate for the formation of saturated and unsaturated six-carbon (C6) volatiles are linoleic acid and α-linolenic acid, respectively. This figure focuses on the formation of the most abundant unsaturated C6-volatiles. For more detailed information on the biosynthesis of GLVs, see recent reviews (Mwenda & Matsui, 2014; ul Hassan *et al.*, 2015). Molecular oxygen is introduced into α-linolenic acid and the resulting hydroperoxide (13(S)-hydroperoxy-9Z,11E,15Z-octadecatrienoic acid; 13-HPOT) is cleaved into a C12-(9Z-traumatins) and a C6-(Z-3-hexenal) compound. 13-HPOT is substrate for different branches of the oxylipin pathway and is thus also used for the formation of jasmonic acid (JA). In damaged leaf tissue, galactolipids can be directly oxygenated without prior cleavage of the free FAs (Nakashima *et al.*, 2013). The resulting lipid hydroperoxide is cleaved into a traumatins containing galactolipid and Z-3-hexenal. Z-3-hexenal is quickly reduced to Z-3-hexenol and can be glycosylated or converted to Z-3-hexenyl acetate. Some plants and insects possess a (3Z):(2E)-enal isomerase that catalyses the formation of E-2-hexenal from Z-3-hexenal. E-2-hexenal can be further converted to its alcohol and acetate or it can be conjugated to glutathione (GSH). GLVs and their derivatives are highlighted in green. LOX, lipoxygenase; HPL, hydroperoxide lyase; ADH, alcohol dehydrogenase; AKR, aldo-keto reductase; ADR, aldehyde reductase; AAT, alcohol acetyl transferase.

of specific lipases that play a role in the liberation of FAs for the GLV pathway (Mwenda & Matsui, 2014). In *Arabidopsis*, the supply of free FAs for the biosynthesis of the plant hormone jasmonic acid (JA), which is formed via another branch of the oxylipin pathway, is regulated by a number of different lipases, all belonging to the class of phospholipases-A1 (Bonaventure, 2014). These lipases can act in different tissues (Ellinger *et al.*, 2010), upon different stimuli (Ellinger & Kubigsteltig, 2010) and at different stages of the stress response. One of them, phospholipase A-1γ1, is involved in the early wound-induced accumulation of JA and 12-oxo-phytodienoic acid (OPDA) (Ellinger *et al.*, 2010). Whether this lipase can also supply substrate for the hydroperoxide lyase (HPL) branch of the oxylipin pathway has not been tested as yet. In *Nicotiana attenuata*, glycerolipase A1 (GLA1) is the major supplier of free FAs for the formation of JA after herbivory, but is not involved in the biosynthesis of GLVs (Bonaventure *et al.*, 2011). So far, only two lipases have been implicated in the formation of GLVs, phospholipase Dα4 and Dα5 (OsPLDα4/5); antisense expression of OsPLDα4/5 in rice reduced not only the concentrations of ALA and JA, but also the release of the GLVs Z-3-hexenal (Z-3-HAL) and Z-3-hexenol (Z-3-HOL) after herbivory or

wounding (Qi *et al.*, 2011). However, this study did not unambiguously show whether PLDα4/5 can directly supply free FAs for the GLV pathway or whether they only indirectly affect GLV concentrations by hampering JA biosynthesis which can feed back on GLV biosynthesis (Wei *et al.*, 2011). From recent research in *Arabidopsis* we know that in damaged tissue GLVs can also be formed without cleavage from membrane lipids (Nakashima *et al.*, 2013) and thus via a lipase-independent pathway. Free or membrane bound FAs are dioxygenated by 13-lipoxygenases (LOX), which are nonheme, iron-containing enzymes catalysing the incorporation of molecular oxygen at the C₁₃ position (Liavonchanka & Feussner, 2006; ul Hassan *et al.*, 2015) of polyunsaturated fatty acids. The resulting 13-hydroperoxy fatty acids (13-HPs) are a common substrate for different enzymes including the two cytochrome P450s (CYP), allene oxide synthase (AOS) and HPL (Gobel & Feussner, 2009; Savchenko *et al.*, 2014). Both AOS and HPL are members of the CYP74 family, an atypical subgroup within the P450 family. The CYP74 family is involved in the metabolism of hydroperoxides, and does not require oxygen or an NADPH-reductase, resulting in high turnover numbers (Noordermeer *et al.*, 2001; Hughes *et al.*, 2009).

Because of the shared substrate, plants need to distinctly regulate GLV and JA biosynthesis. As stated by Mochizuki *et al.* (2016) there are two ways to accomplish this. First, by a distinct spatiotemporal expression of *HPL* and *AOS* (Mwenda *et al.*, 2015), and, second, by expression of pathway-specific LOXs. While some plant species possess pathway-specific LOXs that feed 13-HPs either into the AOS branch alone, leading to the formation of JA, or into the HPL branch, leading to the formation of GLVs (Allmann *et al.*, 2010; Christensen *et al.*, 2013), other plant species have LOXs that can supply substrate to both pathways (Wang *et al.*, 2008; Mochizuki *et al.*, 2016). Competitive substrate flux into the AOS and HP branches has already been reported for several plant species by silencing or overexpression of the HPL/AOS pathways (Halitschke *et al.*, 2004; Liu *et al.*, 2012; Tong *et al.*, 2012; Xin *et al.*, 2014). When fed into the HPL branch, 13-HPs are cleaved into 12 carbon oxoacids, such as 9Z-traumatin, and a C6 compound. Depending on whether ALA or linoleic acid is the substrate for HPL, the resulting C6 compound is either the unsaturated Z-3-HAL or the saturated n-hexanal. Z-3-HAL is relatively unstable and thus rearranges either spontaneously or enzymatically to E-2-HAL (Noordermeer *et al.*, 1999; Allmann & Baldwin, 2010; Kunishima *et al.*, 2016). In previous literature the enzyme responsible for this rearrangement was defined as (3Z):(2E)-enal isomerase. Although, there is some disagreement within the scientific community on this terminology, we will use, for the sake of brevity and uniformity, the term (3Z):(2E)-enal isomerase and the abbreviation HI (hexenal isomerase) for this double bond migration. While extensive research has been done on plant LOXs and HPLs (Andreou & Feussner, 2009; Hughes *et al.*, 2009; Mwenda & Matsui, 2014; ul Hassan *et al.*, 2015), much less is known about the enzymes that are involved in the reduction of C6-aldehydes to their corresponding alcohols. For many years it was assumed that this conversion was catalysed by an alcohol dehydrogenase (ADH; Hatanaka, 1993); however, an *Arabidopsis* mutant lacking ADH activity could not fully abolish the concentrations of Z-3-HOL, indicating that either there is gene redundancy or there are other enzymes involved as well. More recent research revealed that NADPH-dependent aldehyde- or aldo-keto reductases play a role in this conversion (Matsui *et al.*, 2012). Subsequently, alcohols can be modified by the activity of an alcohol acyltransferase (AAT; D'Auria *et al.*, 2007) to their corresponding esters.

III. Meta-analysis

As the production of GLVs has often been reported in studies investigating the headspace of plants after wounding or herbivore attack (Shiojiri *et al.*, 2006a), GLVs have mainly been associated with infestations by herbivorous insects (from now on referred to as herbivores). However, in the last decade, evidence has accumulated that pathogens can also induce the release of GLVs (Piesik *et al.*, 2011; Scala *et al.*, 2013). To obtain a better insight into the production pattern of GLVs, we performed a meta-analysis on 51 studies (163 treatments) (Supporting Information Notes S1), investigating GLV production by plants after herbivory, fungal infection or mechanical damage. Following the procedure described in Rowen & Kaplan (2016), we calculated the treatment

effect (TE) of each study, using Hedges' *g*, which represents the standardized difference of the means between the treatment and the control (Hedges, 1981). Details of the meta-analysis can be found in the Supplementary Information (Methods S1, S2). Possible publication bias was checked by constructing funnel plots (Fig. S1).

IV. The type of stress influences the total amount of GLVs released

Our meta-analysis showed that the amount of GLVs produced differs depending on the type of stress the plant encounters. While plants infected by fungi showed a greater treatment effect (TE = 9.67, $P < 0.001$) than wounded plants (TE = 4.69, $P < 0.001$), the effect after herbivory was the smallest (TE = 1.86, $P < 0.001$) (Fig. 3a; Table S1).

The prompt formation after wounding and the fact that GLVs are formed from plant endogenous components advocate in favour of a role for GLVs as damage-associated molecular patterns (DAMPs) (Duran-Flores & Heil, 2016). However, while both fungal pathogens and herbivores also inflict cellular damage, treatment effects are significantly different from the wounding treatments. This disparity points to an important role for fungi or herbivores in modulating the GLV production. As suggested by Duran-Flores & Heil (2016), plants can make use of a complex mix of 'danger signals' to recognize the nature of the attacker. This mix might consist of endogenous damage signals (DAMPs) and enemy-derived signals, including herbivore-associated molecular patterns (HAMPs) or microbe associated molecular patterns (MAMPs). In the following paragraphs we will discuss the possible role of HAMPs and MAMPs in modulating the wound-induced GLV release.

V. Herbivores can modulate the wound-induced release of GLVs

Plants respond to herbivory with the release of herbivore-induced plant volatiles (HIPVs), which mainly consist of terpenoids, GLVs, aromatic compounds and amino acid derivatives (Clavijo McCormick *et al.*, 2012). While some of the HIPVs are only released upon herbivory and not by unwounded or mechanically wounded plants, others are released in higher quantities compared with wounded plants (Dicke & Baldwin, 2009). To distinguish folivory from pure mechanical damage, plants can recognize the frequency of damage (Mithöfer *et al.*, 2005) or the presence of HAMPs introduced into the wounds during feeding. Several of these HAMPs have been identified from the oral secretions (OSs) of herbivores, including, amongst others, fatty acid-amino acid conjugates (FACs, e.g. volicitin), sulfated fatty acids (caeliferins), and enzymes (e.g. β -glucosidase) (Acevedo *et al.*, 2015). Whereas induction of plant defences is often accomplished via HAMPs, the suppression of induced or constitutive defences is mainly attributed to so-called effector molecules (Kant *et al.*, 2015).

Our meta-analysis suggests that HAMPs do not play a major role in the GLV emission of plants, as the treatment effect for total GLV emission after herbivory is even lower than after mechanical damage. However, it is also possible that effectors are introduced

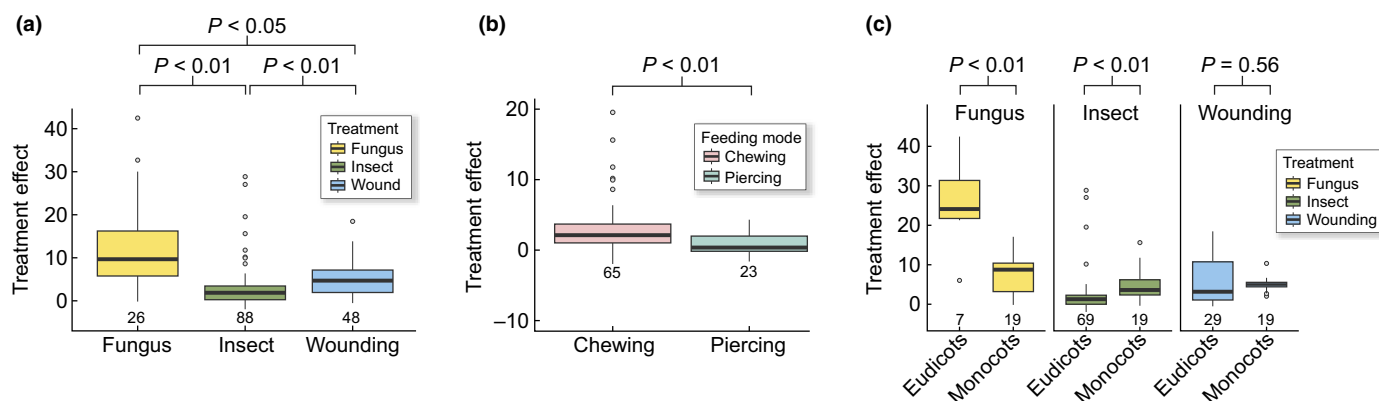


Fig. 3 Effect size on the total production of green leaf volatiles (GLVs) in response to different treatments. (a) The type of treatment has a significant effect on GLV production, with the highest effect for fungal treatments, followed by wounding and herbivory. (b) Herbivory by piercing/sucking herbivores results in lower release concentrations of GLVs compared with chewing herbivores. (c) Treatment effects differ between monocotyledonous and eudicotyledonous plants, with a stronger effect in eudicotyledonous plants after fungal treatment, whereas monocotyledonous plants show a higher effect after herbivory. For easier interpretation of (c), two outliers with an effect size > 20 for the chewers are not shown, but can be seen in (a); the number of studies used to calculate the effect is shown under the boxplots. The central line in the box represents the median value of the treatment effect; the bottom and top lines of the box represent the 25th and 75th percentiles of the treatment effect, respectively. Whiskers represent the minimum and maximum values. Circles represent outliers with a treatment effect that exceeds 1.5 times the interquartile range. Statistical differences are calculated using one-way ANOVA with Welch correction and a *post hoc* Dunnett T3 test for pairwise comparisons ($\alpha < 0.05$).

during herbivory which could mask a possible effect of HAMPs. This yet lower treatment effect of herbivore infestation compared with mechanically wounded plants is quite unexpected, as it is commonly accepted that plants generally increase their GLV emission upon herbivory (Farag *et al.*, 2005; Shiojiri *et al.*, 2006a; Sufang *et al.*, 2013) and that this increase is merely a result of the amount of physical damage that the attacker causes to the plant tissue. As mechanical damage is quite often extensive and applied once, while herbivory causes comparably little damage over a longer period of time, we cannot fully exclude the possibility that the observed differences in treatment effects, compared with the wounding treatment, are simply a shortcoming of our meta-analysis. However, subdividing the herbivory treatment group into real herbivory and simulated herbivory (artificial wounding + OSs) did not lead to a different outcome (Fig. S2). Interestingly, there are only a few studies that tried to disentangle the role of herbivore damage and salivary factors on the plant's GLV emission, which showed either amplification (Mattiacci *et al.*, 1995; Yan & Wang, 2006; Gaquerel *et al.*, 2009) or suppression (D'Auria *et al.*, 2007; Gaquerel *et al.*, 2009; Savchenko *et al.*, 2012; Allmann *et al.*, 2013) of the wound-induced GLV release. This suggests that the assumption – herbivory induced GLV emission results from the inflicted tissue damage during feeding – most likely derived from studies that used intact rather than wounded plants as control treatments.

The examples described later illustrate that the effect of herbivore OSs on the wound-induced release of GLVs clearly depends on many different factors: application of β -glucosidase or OSs from *Pieris brassicae* larvae on wounded cabbage plants caused an increased emission of several GLVs that exceeded the amounts released from purely mechanically damaged plants (Mattiacci *et al.*, 1995). Yan & Wang (2006) demonstrated that regurgitate of two different lepidopteran species, when applied to wounded maize seedlings, caused an amplification of the wound-induced release of

Z-3-HAC. Other GLVs were not affected by the regurgitate treatment. When OSs of *Manduca sexta* larvae were applied to the wounds of *N. attenuata* leaves, these plants emitted higher amounts of several Z-3-hexenyl esters (amongst others Z-3-hexenylpropionate, Z-3-hexenyl, E-2-butenic ester and Z-3-hexenyltigilate) compared with wounded plants (Gaquerel *et al.*, 2009). As these esters were hardly reported in other studies, probably as a result of their very low abundance, they were not included in our meta-analysis. Interestingly, the same study reports a decrease of the wound-induced release of Z-3-HOL when plants were treated with either *M. sexta* OSs or a mix of FACs commonly present in *M. sexta*'s OSs.

Plants can evolve to recognize a pest and initiate direct and indirect defences. This is supported by the fact that OSs of herbivores contain HAMPs which trigger, for example, the accumulation of certain phytohormones (Schmelz *et al.*, 2009) and induce the emission of HIPVs. It is thus not surprising that herbivores can also evolve to suppress these defences; *Arabidopsis* plants which were attacked by *Spodoptera exigua* caterpillars released much smaller quantities of Z-3-HOL and Z-3-HAC compared with mechanically wounded plants (D'Auria *et al.*, 2007). Similar results were obtained when *Arabidopsis* leaves were treated with *P. rapae* OSs (Savchenko *et al.*, 2012). This OS-induced suppression in the wound-induced GLV emission was regulated both transcriptionally and post-transcriptionally, and was restricted to the HPL branch of the oxylipin pathway; JA concentrations were still up-regulated in these plants upon OS treatment (Savchenko *et al.*, 2012). In the wild tobacco plant, *N. attenuata*, the wound-induced increase in HPL transcript abundances was suppressed by application of *M. sexta* OSs onto wounded leaves. Surprisingly, this suppression was only detectable on the transcriptional and not on the metabolic level: GLV emission of OS-treated tobacco plants was even higher than that of wound-induced plants (Halitschke *et al.*, 2004). The authors

argued that this discrepancy – low *HPL* transcripts and high GLV emission – probably results from the fact that GLV biosynthesis is not controlled transcriptionally but rather by substrate flux, and that the release is limited by the supply of fatty acid 13-HPs that are readily metabolized by a high constitutive *HPL* activity that has been found in wild tobacco (Ziegler *et al.*, 2001). While evidence accumulates that the rate-limiting step is indeed upstream of *HPL* (Shiojiri *et al.*, 2006a; Hughes *et al.*, 2009), the time frame that was chosen for this study to measure GLVs was too short to abandon the possibility that total GLV emissions might be suppressed by OS application at a later time point. While volatiles were only measured within the first 20 min after treatment, the transcriptional analysis showed that suppression of *HPL* started only 1 h after OS treatment (Halitschke *et al.*, 2004). Additionally, volatile measurements on wild tobacco plants that were taken 1–13 h after OS treatment did show that Z-3-HOL as well as 1-hexanol emissions were decreased compared with wounded plants (Gaque-rel *et al.*, 2009). This is also apparent from our meta-analysis as we see a greater treatment effect at earlier time points for herbivores compared with wounding, but which attenuates at later time points (Fig. 4). Recent research reveals that light and other (a)biotic factors also play a role in the fine-tuning of the OS-induced modulation of GLV emission; OS-induced suppression in total GLV emission under laboratory low light conditions vanished when plants were grown in the field and thus encountered other (a)biotic stress factors as well (Allmann *et al.*, 2013).

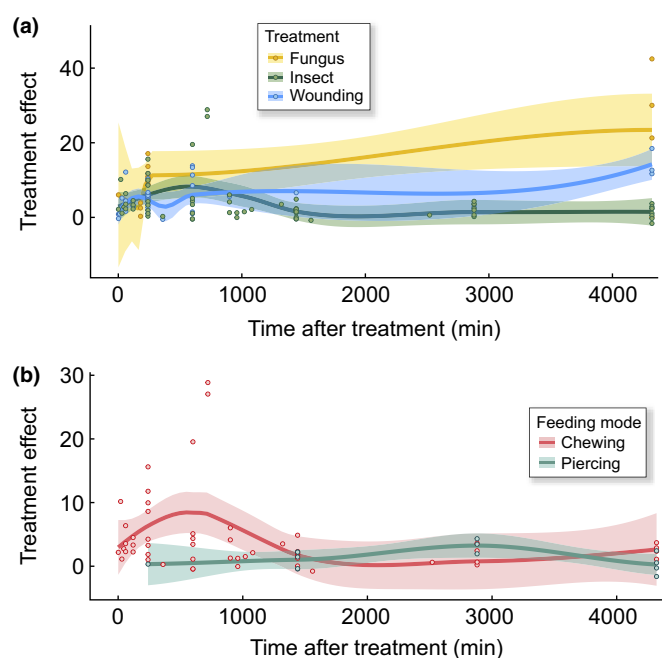


Fig. 4 Time effect of different treatments on green leaf volatile (GLV) production. (a) Treatment effects at early time points are similar across the different treatments. However, at later time points, fungal treatments induce a greater effect on GLV production compared with the other treatments. Additionally, insects induce the lowest effect on GLV production. (b) Within the insect treatment, insects with a piercing/sucking mode do not induce, or only slightly induce, GLV production at early time points, in contrast to chewing insects, which induce a greater effect that levels off at later time points. Lines represent locally weighted polynomial regression (LOESS), and shaded areas represent the 95% confidence interval of the LOESS line.

Suppression of direct and indirect defences has been shown for chewing as well as for piercing/sucking herbivores, such as mites, aphids and whiteflies (Kant *et al.*, 2015). Our meta-analysis demonstrated a difference in treatment effect depending on the feeding mode of the herbivore; while the treatment effect of chewing insects was 2.1 ($P < 0.001$), piercing/sucking herbivores showed a much smaller effect (TE = 0.36, $P < 0.05$) (Fig. 3b). The lower GLV production upon infestation by piercing insects compared with chewing insects was consistent throughout a bigger time frame (Fig. 4).

The difference in the treatment effect between the two feeding guilds, as described in our meta-analysis, might be attributed to differences in their ability to suppress or avoid recognition and defences. However, the most obvious explanation is that piercing/sucking herbivores simply cause less damage to the plant, thereby avoiding an increased release of GLVs. It is also possible that the difference in total GLV emission is caused by the activation of different defence pathways employed by plants. In general, but with an increasing number of exceptions (Kant *et al.*, 2015), piercing/sucking herbivores induce the salicylic acid (SA) pathway, which can have an antagonizing effect on JA-induced defences, whereas chewing herbivores mainly activate the JA pathway and subsequent defences (Zarate *et al.*, 2007; Kant *et al.*, 2015). Because an increase in GLV emission often coincides with a higher JA-associated defence (Scala *et al.*, 2013), the induction of JA-mediated defences upon herbivore attack by chewers might explain the increased production of GLVs. Interestingly, tobacco plants with reduced transcript abundances of *NaLOX3* – the LOX that specifically supplies substrate to the JA pathway – produced not only lower amounts of JA compared with WT plants but had also reduced *HPL* transcript abundances (Halitschke & Baldwin, 2003; Allmann *et al.*, 2010). However, our meta-analysis does not answer the question of whether piercing/sucking herbivores are better at suppressing GLV emissions, either directly or indirectly by inducing hormonal defence pathways that counteract the GLV-inducing JA pathway, or whether they simply cause less damage to the plant with their feeding style and thus induce lower levels of GLVs compared with chewers. In a recent meta-analysis on induced plant volatiles performed by Rowen & Kaplan (2016), piercing/sucking herbivores also showed a smaller treatment effect on GLV production, compared with chewers, which the authors attributed to differences in induced defence pathways, rather than differences in cellular damage caused by feeding. There are only very few studies that directly compare the role of different feeding guilds in inducing GLV emissions: in *Vicia faba* plants, pea aphid feeding did not induce GLV emission, but when plants were coinfecting with *S. exigua* caterpillars, volatile emissions, including Z-3-HOL and Z-3-HAC, were decreased compared with those plants that had been infested with caterpillars only (Schwartzberg *et al.*, 2011). This indicates that aphid-derived effector molecules play a role in suppressing herbivory-induced GLV emissions.

In the case of wound-induced GLV suppression by herbivores it remains an open question whether it is the plant that evolved to recognize HAMPs or whether it is the herbivore that evolved to produce effectors in order to actively suppress wound-induced GLV emissions. From the plant's perspective, GLVs have been

shown to increase the attractiveness of plants to herbivores (Halitschke *et al.*, 2004; Han *et al.*, 2012) and 'getting off the radar' (Halitschke *et al.*, 2008) could help the plant to protect itself against herbivore attack. Additionally, GLVs have been shown to serve as feeding stimulant by increasing the consumption rate of lepidopteran caterpillars, and decreasing the amount of GLVs might be a good strategy to delay larval growth (Halitschke *et al.*, 2004). On the other hand, GLVs are also known to attract predators and parasitoids to the herbivore-infested plants (Shiojiri *et al.*, 2006b; Allmann & Baldwin, 2010; Shimoda, 2010) and 'getting off the radar' could provide a degree of protection for the herbivore as well. In tomato, glycosylation of exogenous Z-3-HOL to Z-3-hexenylvicinioside had a direct toxic effect on the common cutworm *Spodoptera litura* (Sugimoto *et al.*, 2014). Thus, suppression of GLV production might be a measure for herbivores to reduce the amount of toxic GLV metabolites.

VI. Fungal infection greatly induces GLV production

While it is clear that GLVs mediate plant–insect interactions and that both plants and insects can benefit from altering the GLV production, it is more of a mystery how fungi may benefit from inducing GLV production. The meta-analysis showed that fungal infection caused the highest treatment effect compared with wounding and herbivory (Fig. 3a). It remains unclear whether the high GLV production is an active response by the plant following recognition of fungal MAMPs or a strategy by the pathogen to induce GLV production to promote virulence, or whether GLV production is merely a by-product of cellular oxidative damage. In addition, the lifestyle of the fungus most likely also influences GLV production. Whereas biotrophic fungi thrive on living cells, necrotrophic fungi kill the host cell, thereby causing more cellular damage, which also may result in increased GLV production.

Unlike herbivores, fungal pathogens are restricted within the plant tissue, forcing them to employ sophisticated mechanisms to thrive on these plant cells. Production of phytotoxins (Möbius & Hertweck, 2009) as well as hijacking the plant hormone defence network are known strategies to successfully infect plant tissue (Maor & Shirasu, 2005; Yan & Xie, 2015). Some of these toxins are known to cause oxidative stress within the cell, which may lead to membrane damage resulting in substrate availability for GLV biosynthesis (Möbius & Hertweck, 2009). However, the significantly larger effect compared with wounding suggests that substrate availability does not fully account for the increase in GLV production after infection. It thus seems more likely that fungal effectors are involved in the increased GLV emission. To our knowledge, no fungal effectors have been described which attenuate or enhance GLV production. Notwithstanding this, Blümke *et al.* (2014) showed that the pathogen *Fusarium graminearum* releases the effector lipase FGL1 to inhibit callose deposition in spikelets of wheat, thus enabling further colonization. The $\Delta fgl1$ mutant showed, together with reduced virulence, increased callose depositions and reduced concentrations of the free FAs linoleic acid and ALA. In addition, exogenously applied free FA restored the virulence of $\Delta fgl1$ to WT values, showing that free FAs have an important role in the suppression of the innate

immunity related to callose biosynthesis. Thus, it is tempting to speculate that FGL1 can increase the availability of free FAs, which could provide more substrate for GLV biosynthesis (Fig. 2). Recently a link between GLVs and the release of free FA was found; Li *et al.* (2016) reported a transient increase in ALA after treating *Zea mays* and *Solanum lycopersicum* with Z-3-HOL. They also found evidence of a priming effect of free FA for enhanced JA accumulation in distal plant tissue of corn seedlings which could not be attributed to the increased substrate availability of ALA for JA biosynthesis. Together, these papers provide an indication that fungi could interfere with primary plant defences (callose deposition) by influencing GLV biosynthesis and the subsequent free FA release.

The high GLV production after fungal infection can also be attributed to the plant hormonal defence response. Similar to insect infestations, fungal infection induces changes in defence hormone concentrations. Because of the many antagonistic and synergistic signalling pathways between different defence hormones, this leads to a whole rewiring of the plant hormone network, allowing the plant to finely regulate its defences to fend off fungal infection (Pieterse *et al.*, 2012). Our meta-analysis predominantly contains studies on necrotrophic fungi (e.g. *Fusarium* spp., *Botrytis cinerea*), which have been shown to increase JA concentrations in plants after infection (Ding *et al.*, 2011; El Oirdi *et al.*, 2011). It has been reported repeatedly that JA and GLVs are mutually influenced by each other (Kessler & Baldwin, 2001; Engelberth *et al.*, 2004, 2013; Bruinsma *et al.*, 2009; Wei *et al.*, 2011), which may account for the high induction of GLVs after fungal infection. It would be interesting to investigate the effect of biotrophic fungi on GLV production. Biotrophs might benefit more from a reduction in GLVs than do necrotrophs. Additionally, research including biotrophs might shed light on the question of whether the GLV induction is primarily MAMP- or effector-driven or whether the amount of damage caused by biotrophic or necrotrophic fungi is the main driver of GLV production.

An active role for fungi in inducing GLV production is somewhat counterintuitive, because GLVs function as defensive compounds by exerting a direct negative effect against pathogenic fungi or by inducing and priming plant defences (Kishimoto *et al.*, 2005, 2006b; Ameye *et al.*, 2015). For example, Kishimoto *et al.* (2006a) found that exposing *Arabidopsis thaliana* to the GLVs E-2-HAL and Z-3-HAL enhanced resistance against the necrotrophic fungus *B. cinerea*. They attributed this enhanced resistance to increased lignification and higher transcripts of PDF1.2 and PR3, a chitinase. Further research by the same group additionally revealed that aforementioned GLVs possess fungicidal activity by inhibiting conidia germination and hyphae growth (Kishimoto *et al.*, 2008). The antifungal activity for C6 aldehydes has also been shown against other fungi such as *Alternaria alternata*, *B. cinerea* (Hamilton-Kemp *et al.*, 1992) and *Aspergillus flavus* (De Lucca *et al.*, 2011). On the other hand, fungi might induce GLV production to influence another trophic level. Induction of GLV production by fungi might benefit fungal infection and conidia dispersion by attracting herbivores which serve as vectors (Kluth *et al.*, 2002). Indeed, GLVs produced by infected plants are known to attract aphids (Webster & Cardé, 2017), which have recently been shown

to promote fungal disease in wheat (Drakulic *et al.*, 2015, 2016; De Zutter *et al.*, 2017).

However, fungi have often been shown to exploit the antagonistic signalling between defence hormones and, as such, hijack the plant defence machinery; in other words, fungi do not suppress plant defences but manipulate the plant to invest in defence responses which are not effective to the pathogen in play or which even induce susceptibility (Verhage *et al.*, 2010). This interference in the plant's metabolism is known to take place at several levels comprising both primary and secondary metabolism (Walton, 1996; Möbius & Hertweck, 2009). Thus, an increased defence response and concomitant increased GLV release does not necessarily mean a successful control of the pathogen by the plant. Indeed, the outcome of the GLV production during fungal infection is probably the combined effect of plant defence and fungal interference with that defence. As every fungal pathogen has its own arsenal of effectors (e.g. toxins), each interfering with plant defence in a specific way, one might expect that the GLV release in these plant fungal interactions is very variable. Our meta-analysis supports this hypothesis nicely: the variation of the treatment effect after fungal infection is large compared with the treatment effect after insect infestation or wounding (Fig. 3a). This high variability is consistent in a broader time frame (Fig. 4a).

Similar to herbivores, fungi are able to locate hosts using chemical cues; hyphae can orient their growth towards chemical gradients (Brand & Gow, 2009). Recent research showed that the soilborne fungus *Fusarium oxysporum* can quickly reorient its growth in response to sugars, amino acids and root exudates (Turrà *et al.*, 2015). Because of the conserved nature of GLV production it may be possible that pathogenic fungi have evolved to utilize these signals to orient themselves away from (e.g. biotrophs) or towards (e.g. necrotrophs) these volatile cues. Investigating the interaction between fungal chemotropism and plant-derived stress signals such as GLVs provides an exciting field of research (Turrà *et al.*, 2016).

Until recently, GLVs have not been associated with fungal or bacterial infection. However, evidence is accumulating that GLVs also play an important role in plant–pathogen interaction. Still, compared with plant–insect research, there remains a knowledge gap. In order to shed more light on the role of GLVs in plant–pathogen interactions, more research on drivers of GLV biosynthesis after infection is mandatory.

VII. Monocots and eudicots respond differentially to different types of stress

Our meta-analysis demonstrated an interaction between the type of stress the plant undergoes and whether the plant was mono- or eudicotyledonous (Table S2). Remarkably, for infections with fungi, the treatment effect (i.e. the standardized effect of a certain treatment on GLV production) was higher in dicots than in monocots (24.10 vs 8.76, respectively, $P=0.027$), whereas for infestations with herbivores, monocots showed a higher treatment effect compared with eudicots (3.61 vs 1.26, respectively, $P<0.001$) (Fig. 3c). No difference between monocots or eudicots was found in the wounding treatment. To our knowledge, this is the first time that this discrepancy has been reported, leaving us to

speculate on the underlying mechanism. The difference in plant response to biotic stress between monocots and eudicots has already been demonstrated for plant defence hormones. While in dicots, there is a dichotomous model between SA (resistance against biotrophic pathogens) and JA (resistance against necrotrophs and herbivores), in the monocot crop rice, there are multiple cases in which JA provides resistance against both biotrophic and necrotrophic pathogens (De Vleeschauwer *et al.*, 2014). As there is currently an absence of knowledge regarding the mechanism by which GLVs are perceived by the cell and how the signal is transduced, we cannot pinpoint whether the difference between monocots and eudicots can be situated there or whether it lies further downstream. Another hypothesis is that different MAMPs and HAMPs (e.g. chitin for fungal infection and FACs for herbivores) induce different responses in monocots and eudicots. Substrate competition does not seem to explain the difference between monocots and eudicots in general. In eudicots, HPL and AOS colocalize within the chloroplast (Froehlich *et al.*, 2001; Halitschke *et al.*, 2004; Farmaki *et al.*, 2007), facilitating substrate competition for 13-HPs. However, in monocots there is no consistency: in rice negative crosstalk between HPL and AOS has been reported (Liu *et al.*, 2012; Tong *et al.*, 2012) but in maize the JA producing LOX8 is localized in the chloroplast, while the GLV producing LOX10 is localized to nonchloroplast organelles, thus avoiding substrate competition (Christensen *et al.*, 2013). Whether the difference in effect between monocots and dicots can be explained by differences in hormone signalling or whether effectors or substrate competition play a role remains to be elucidated.

VIII. The type of stress does not influence the proportion of GLVs per chemical class

Our meta-analysis provides convincing proof that plants increasingly release GLVs in response to biotic attackers. To determine whether the type of stress additionally has an influence on the composition of GLVs released from the plant, we subdivided the volatiles into three groups according to their chemical classes: aldehydes, alcohols and acetates. The meta-analysis revealed that wounding, herbivory or fungal infection did not cause a change in the relative composition of GLVs, compared with the control treatments, with the exception of the aldehydes after fungal infection (Table 1). The general absence of a shift in the composition is most probably a result of the enzyme reaction kinetics of GLV biosynthesis; overexpression of a bell pepper *HPL* in *Arabidopsis* (Ecotype No-0) did not lead to a constitutive increase in GLV emission compared with WT plants, which indicates that the limiting factor, at least under control conditions, must be upstream of HPL (Shiojiri *et al.*, 2006a). Similar conclusions have been drawn by other researchers (Hughes *et al.*, 2009) some of whom suggested that lipases (Matsui *et al.*, 2000) are the rate-limiting step for GLV production. The 13-HPs which are produced by LOX (Fig. 2) are most probably immediately cleaved by a highly active HPL, as has been proposed for *N. attenuata* (Ziegler *et al.*, 2001). Wounded *Arabidopsis* leaves (No-0) produced mainly Z-3-HOL while the concentrations of Z-3- and E-2-HAL were comparable to intact leaves. This lack of an increase in

Table 1 The relative composition of green leaf volatiles (GLVs) per chemical class produced in plants after different treatments

Treatment	Group	Median (Q1–Q3 quantile)		P	n
		Before treatment	After treatment		
Fungus	Aldehydes	0.450 (0.276–0.583)	0.514 (0.474–0.576)	0.006	22
	Alcohols	0.217 (0.170–0.402)	0.177 (0.146–0.327)	0.115	22
	Acetates	0.273 (0.188–0.309)	0.293 (0.132–0.344)	0.320	22
Insect	Aldehydes	0.191 (0.146–0.481)	0.375 (0.063–0.506)	0.375	25
	Alcohols	0.296 (0.159–0.387)	0.200 (0.143–0.308)	0.109	25
	Acetates	0.385 (0.217–0.621)	0.470 (0.278–0.647)	0.347	25
Wounding	Aldehydes	0.450 (0.190–0.593)	0.488 (0.404–0.532)	0.960	12
	Alcohols	0.176 (0.124–0.479)	0.216 (0.127–0.331)	0.464	12
	Acetates	0.282 (0.240–0.335)	0.338 (0.262–0.405)	0.298	12
Chewing	Aldehydes	0.302 (0.159–0.506)	0.441 (0.187–0.525)	0.200	19
	Alcohols	0.267 (0.156–0.367)	0.200 (0.151–0.269)	0.187	19
	Acetates	0.316 (0.212–0.595)	0.421 (0.252–0.569)	0.646	19
Piercing	Aldehydes	0.161 (0.013–0.178)	0.117 (0.007–0.182)	0.706	6
	Alcohols	0.355 (0.237–0.415)	0.259 (0.127–0.315)	0.385	6
	Acetates	0.540 (0.385–0.749)	0.667 (0.616–0.691)	0.383	6

To calculate relative compositions of GLVs, we converted data from the studies used in the meta-analysis (Supporting Information Notes S1). Statistical differences were calculated using a paired *t*-test ($\alpha < 0.05$). Median values are shown in the table with the Q1 and Q3 quantile between parentheses.

aldehydes suggests that the turnover rate, and thus either the activity or abundance of ADH, AKR or ADR, must be very high as well (Matsui *et al.*, 2012). The fast production (Turlings *et al.*, 1998) and subsequent conversion of *Z*-3-HAL to its alcohols and esters have been described in several studies (Fall *et al.*, 1999; D'Auria *et al.*, 2007; Maja *et al.*, 2014). In wounded aspen leaves *Z*-3-HAL production peaked 5 min after wounding, declined rapidly and coincided with an increase of hexenyl alcohols and acetates between which the ratios stayed constant (Fall *et al.*, 1999). Our meta-analysis suggests that after an initial burst of hexenals, other GLVs are quickly formed and reach constant ratios regardless of the type of biotic stress.

The fact that GLVs are rapidly released from damaged plant tissue makes them an interesting group of semiochemicals, which could provide immediate information about the exact location of an attacking herbivore. However, the question arises: is the information encoded in the GLV bouquet also reliable and specific enough for prey searching carnivores if the composition does not change, as suggested in our meta-analysis? Previous research has shown that quantitative changes in single GLV compounds or a set of GLVs can clearly influence the behaviour of predators and parasitoids (Kessler & Baldwin, 2001; Shiojiri *et al.*, 2006a; Chehab *et al.*, 2008).

While a change in the total amount of GLVs with no change in the composition most probably encodes for an insect only limited information, that is, the amount of feeding damage rather than the nature of the attacker or the host plant, those quantitative changes in the GLV bouquet do affect the ratio between GLVs and other plant volatiles. As insects often rely on a whole volatile blend rather than on single compounds for host plant selection (Bruce & Pickett, 2011) these relative changes compared with other groups of plant volatiles can suffice for insects to make informed choices.

While the nonspecificity in the GLV signal might be an impairment for a host-searching insect, a simple increase in total GLV emission upon, for example, damage should be sufficient to

serve as a damage signal for the plant itself to either induce or prime defence responses (Duran-Flores & Heil, 2016).

IX. The type of stress does influence the isomeric ratio within each chemical class

While at first glance it seems as if herbivore-specific information cannot be conferred via the release of GLVs, recent studies revealed that both plants (Kunishima *et al.*, 2016) and insects (Allmann & Baldwin, 2010) are able to fine-tune the plant's GLV composition; in wild tobacco, herbivory by *M. sexta* caterpillars or the application of their OSs to leaf wounds increased the conversion from *Z*-3- to *E*-2-GLVs. This decrease in the *Z*-3 : *E*-2 ratio tripled the foraging efficiency of the generalist predator *Geocoris* spp. in nature (Allmann & Baldwin, 2010). In this specific case it is not the plant that is responsible for the change in the GLV profile, but an enzyme present in the OSs of the caterpillar. However, plants are also able to convert *Z*-3-HAL to its *E*-2-isomer by themselves: (3*Z*):(2*E*)-enal isomerase (HI) activity has been detected in the crude extracts of some plant species (Phillips *et al.*, 1979; Takamura & Gardner, 1996; Noordermeer *et al.*, 1999) and its sequence has recently been identified from red bell pepper (Kunishima *et al.*, 2016).

To determine whether conversion from *Z*-3- to *E*-2-hexenal is a general phenomenon in plants and whether the type of stress has an influence on this conversion, we compared *Z*-3 : *E*-2 ratios of aldehydes and alcohols after fungal infection or herbivory. The wounding treatment and hexenyl acetates were not included in this analysis owing to the low number of studies that reported both isomeric forms. Our meta-analysis revealed that *Z*-3 : *E*-2 ratios significantly increased after fungal treatment (i.e. low conversion to *E*-2-GLVs) while herbivory did not cause a change in the ratio compared with control treatments (i.e. high conversion to *E*-2-GLVs) (Fig. 5). Plant HIs are generally highly expressed in ripe fruits (paprika and cucumber, Phillips *et al.*, 1979; Kunishima

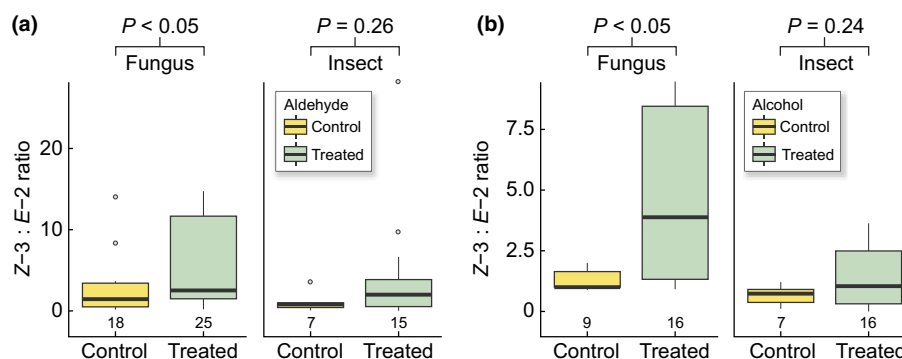


Fig. 5 Z-3 to E-2 ratio of green leaf volatile (GLV) isomers of (a) aldehydes and (b) alcohols. The meta-analysis revealed an increase in the (Z-3) : (E-2)-ratio after fungal treatments, whereas the (Z-3) : (E-2) ratio after herbivory did not significantly differ compared with intact plants. The central line in the box represents the median value of the Z-3 : E-2 ratio; the bottom and top lines of the box represent the 25th and 75th percentiles, respectively. Whiskers represent the minimum and maximum values. Circles represent outliers with a value of the Z-3 : E-2 ratio treatment effect that exceeds 1.5 times the interquartile range. Statistical significance is calculated using one-way ANOVA with Welch correction and a *post hoc* Dunnett T3 test ($\alpha < 0.05$).

et al., 2016) and potato sprouts but show rather low constitutive expression levels in leaves. Wound-induced E-2-HAL emissions are increased during the night (De Moraes *et al.*, 2001; Allmann *et al.*, 2013) and mechanical damage increases HI transcript abundances (Kunishima *et al.*, 2016). This shows that plant HIs can be up-regulated under certain circumstances; however, this is apparently not the case upon fungal infection (this meta-analysis). This is rather unexpected as E-2-HAL is well known for its antimicrobial properties and has been reported, at least in some cases, to be even more effective than Z-3-HAL (Kishimoto *et al.*, 2005; Prost *et al.*, 2005). Future research could reveal whether fungi are able to manipulate the plant to produce less E-2-HAL.

Herbivory, on the other hand, clearly increases the conversion from Z-3- to E-2-HAL (Fig. 5). However, whether this general conversion is caused mainly by wound-inducible plant HIs or by insect-derived HIs introduced into the wound during feeding is still unanswered. While an increased conversion has been observed with OSs or herbivory by various lepidopteran species (Turlings & Wäckers, 2004; Allmann & Baldwin, 2010), how widespread the occurrence of such HIs in herbivores still needs to be tested.

The shift from Z-3- to E-2-GLVs seems to be an important way to bring specificity to the GLV blend. However, it is still not clear what is the main function of this increased conversion. What is the evolutionary origin of HIs? Why do plants as well as insects possess HI activity? This might not be an easy question to answer, as it has been shown that an increase in E-2-GLVs can be beneficial for both plants and insects by attracting natural enemies of the herbivores (Allmann & Baldwin, 2010) and by informing gravid female moths about the appropriateness of the host plant (Allmann *et al.*, 2013). However, there are other, not mutually exclusive, hypotheses regarding why plants and insects might increase the conversion rate from Z-3-HAL to E-2-HAL and in the following we will elaborate on these.

First, herbivores and plants may actively produce E-2-HAL to protect themselves from pathogens; GLVs and especially C6-aldehydes have antimicrobial properties against bacteria (Nakamura & Hatanaka, 2002) and fungi (Hamilton-Kemp *et al.*, 1992). There are only few studies that directly compare the antimicrobial

properties of Z-3-HAL and E-2-HAL. While in some cases the two alkenals did not differ in their bacteriostatic effect (Nakamura & Hatanaka, 2002) or their antifungal activity (Tajul *et al.*, 2012), others reported higher antimicrobial activities of E-2-HAL compared with Z-3-HAL (Kishimoto *et al.*, 2005; Prost *et al.*, 2005).

Second, herbivores may actively convert Z-3-HAL to E-2-HAL to boost their innate immune system/deal better with oxidative stress: E-2-HAL is a reactive electrophile species (RES; Alméras *et al.*, 2003) as it possesses an α,β -unsaturated carbonyl group which shows high reactivity with nucleophilic atoms and can cause protein carbonylation (Farmer & Davoine, 2007; Mueller & Berger, 2009). RES-mediated carbonylation often leads to the inactivation of proteins (Liebler, 2008); however, it can also result in the activation of Cap'n'collar (Cnc) proteins which play an important role in regulating cellular defences against oxidative stress (Nguyen *et al.*, 2009). The most studied Cnc transcription factor is the vertebrate homolog Nrf2 (Sykietis & Bohmann, 2010) and an ortholog, called CncC, has been identified for invertebrates (*Drosophila*; Sykietis & Bohmann, 2008). Further research will reveal whether plants have a similar system to cope with oxidative stress.

Third, plants may actively produce E-2-HAL to induce or prime plant defences in neighbouring plants or adjacent leaves of the same plant. This has been described for several GLVs (see the following section) but only a few studies have directly compared the efficiency of Z-3- and E-2-GLVs in inducing plant defence responses. While in maize seedlings Z-3-GLVs elicited a stronger response than E-2-GLVs (Ruther & Fürstenau, 2005), tomato plants showed an opposite trend by releasing higher amounts of monoterpenes when exposed to E-2-HAL compared with Z-3-HAL (Farag & Paré, 2002). In any case, E-2-HAL has been shown to serve as a potent inducer of plant defences (Zeringue, 1992; Bate & Rothstein, 1998; Kessler *et al.*, 2006) and several E-2-HAL-specific marker genes have recently been identified in *Arabidopsis* (Mirabella *et al.*, 2015). Clearly, much more work is needed to understand whether compositional changes in the GLV bouquet are recognized by plants and used to modulate the induction or priming of plant defences.

X. GLVs: from signal perception to signal transduction

Volatiles can be used by insects to make informed choices about mating partners, habitats and appropriate host plants as food source or oviposition site (Dicke & Baldwin, 2009; Gadenne *et al.*, 2016), but also by plants to induce or prime defence responses (Kant *et al.*, 2009). Whereas the effect of GLVs on their interactors has been discussed within this review, this section will specifically focus on the functionality of GLVs for plants and discuss the recent knowledge of GLV perception and transduction in plants.

The idea that signalling between plants might be mediated by volatiles to activate defence responses emerged in the early 1980s (Baldwin & Schultz, 1983; Rhoades, 1983). Although controversial at that time, many studies have followed since, confirming that volatiles can initiate plant defence responses (Kant *et al.*, 2009; Karban *et al.*, 2014). Treating plants with GLVs can induce the expression of several defence-related genes and downstream metabolites (Zeringue, 1992; Bate & Rothstein, 1998; Engelberth *et al.*, 2013), including a subsequent release of BVOCs (Farag & Paré, 2002) or the secretion of extrafloral nectar (Kost & Heil, 2006). However, plants do not always directly up-regulate their defences when exposed to plant volatiles. They can also be alerted by these volatiles, which enables them to induce their defences more rapidly and/or more effectively at the actual time of herbivore or pathogen attack. This form of alertness is called priming and it has the advantage that it does not involve a significant fitness penalty for the plant (Martinez-Medina *et al.*, 2016). A pioneering study by Engelberth *et al.* (2004) showed that exposure of maize plants to pure GLVs increased JA production and the release of several terpenoids after challenge with crude regurgitant from *S. exigua*. Since then, several other studies have followed and confirmed the priming effect of GLVs on plants upon herbivory (Kost & Heil, 2006; Engelberth *et al.*, 2007, 2013; Frost *et al.*, 2008; Li *et al.*, 2016). Recently, the priming potential of GLVs has also been shown to act against pathogens. Exposing wheat to Z-3-HAC primed plants for enhanced defence against the pathogenic fungi *F. graminearum* by increasing JA-dependent signalling (Ameye *et al.*, 2015). Even though in the latter study, wheat seedlings were exposed to a high concentration of Z-3-HAC (0.11 mM), aerial Z-3-HAC concentration declined very rapidly to previously reported concentrations (Wenda-Piesik *et al.*, 2010; Piesik *et al.*, 2011). As stated by Matsui *et al.* (2012), aqueous concentrations of GLVs within cells can reach values up to 1 mM. Thus, using a high concentration for a short period of time may actually mimic GLV concentrations in damaged cells. Plants can eavesdrop on the volatile signals coming from their neighbours, but plant signalling by GLVs is more likely to occur within one plant and might serve to overcome vascular constraints or augment vascular systemic signals (Heil & Ton, 2008). In the case of intra-plant signalling GLVs should be defined as DAMPs as proposed by Duran-Flores & Heil (2016).

The observation that plants can use volatile information coming from neighbouring plants or adjacent leaves raises questions about the perception and signal transduction of GLVs. Whereas in recent years an increasing amount of knowledge has been gained with regard to the molecular and cellular processes of odorant detection

in insect antennae and the primary processing of olfactory signals in the brain (Carey & Carlson, 2011), knowledge regarding volatile perception by plants is still in its infancy (Heil, 2014) and a volatile receptor has so far only been identified for ethylene (Schaller & Bleecker, 1995). In insects, receptor proteins are key elements for the recognition and discrimination of plant volatiles (Leal, 2013); different types of GLVs can cause various types of response in insects, at both the physiological (Hansson *et al.*, 1999; Røstelien *et al.*, 2005; Allmann *et al.*, 2013) and behavioural levels (Reinecke *et al.*, 2005; Allmann *et al.*, 2013). In plants, such specificity has been shown in some but not all cases: while structurally different GLVs (Z-3-HOL, Z-3-HAL and E-2-HAL) can have different efficacies in activating monoterpene emission in tomato (Farag & Paré, 2002), neither configuration nor the position of the double bond of GLVs were important for the induction of extrafloral nectar in lima beans (Heil *et al.*, 2008).

In order to transfer GLV-encoded information into plant cells, GLVs need to reach the plasma membrane or other organelles of the plant cell, via either active or passive transport. Plant volatiles are most likely taken up by the plant via their stomata or by adsorption through the leaf surface (Matsui, 2016). Still, before reaching the plasma membrane they have to cross the cuticle and the cell wall. How this occurs is still unknown, but as a result of their lipophilic character, GLVs can 'dissolve' in the plasma membrane as suggested by Heil (2014), but they can also reach the cytosol where they are further metabolized by the plant cell (Fig. 6; Farag & Paré, 2002; Matsui *et al.*, 2012). Reported metabolism processes of GLVs include glutathionylation of E-2-HAL (Fig. 6, IV; Davoine *et al.*, 2006; Mirabella *et al.*, 2008) and the glycosylation of Z-3-HOL (Sugimoto *et al.*, 2014). Early events in leaves upon GLV exposure include plasma membrane potential (V_m) depolarization and an increase in $[Ca^{2+}]_{cyt}$. Exposure of tomato plants to several GLVs triggered within seconds a depolarization of mesophyll cells (Fig. 6, 1) and within minutes an increase in cytosolic calcium $[Ca^{2+}]_{cyt}$ (Zebelo *et al.*, 2012). Similar results have been found in *Arabidopsis*; exposure of leaves to E-2-HAL and E-2-HOL induced a fast and transient increase in $[Ca^{2+}]_{cyt}$ (Asai *et al.*, 2009). This increase can be caused by the activation of a depolarization-activated calcium channel (DACC) (Fig. 6, 2a), a ROS-activated calcium channel (Fig. 6, 2b) or via the release of Ca^{2+} from vacuoles (Fig. 6, 2c; Swarbreck *et al.*, 2013). While activation of DACCs is most likely triggered by many if not all GLVs (Zebelo *et al.*, 2012), an increase in $[Ca^{2+}]_{cyt}$ via ROS-activated calcium channels seems to be more specific for RES-type GLVs, such as E-2-HAL (Asai *et al.*, 2009).

Some of the downstream molecular players in GLV signalling have recently been elucidated; exposure of maize leaves to Z-3-HOL for either 20 or 60 min increased the transcript abundance of several genes involved in signalling, including transcription factors, genes related to phosphorylation, calcium (Fig. 6, 3) and lipid signalling. While transcriptional regulators accounted for almost 50% of the differentially regulated genes 20 min after GLV exposure, they represented only a minor group of 10–15% after 60 min. Interestingly, there was hardly any overlap in regulated genes between early and late time points (Engelberth *et al.*, 2013). Other early signalling responses include (Fig. 6, 4) the

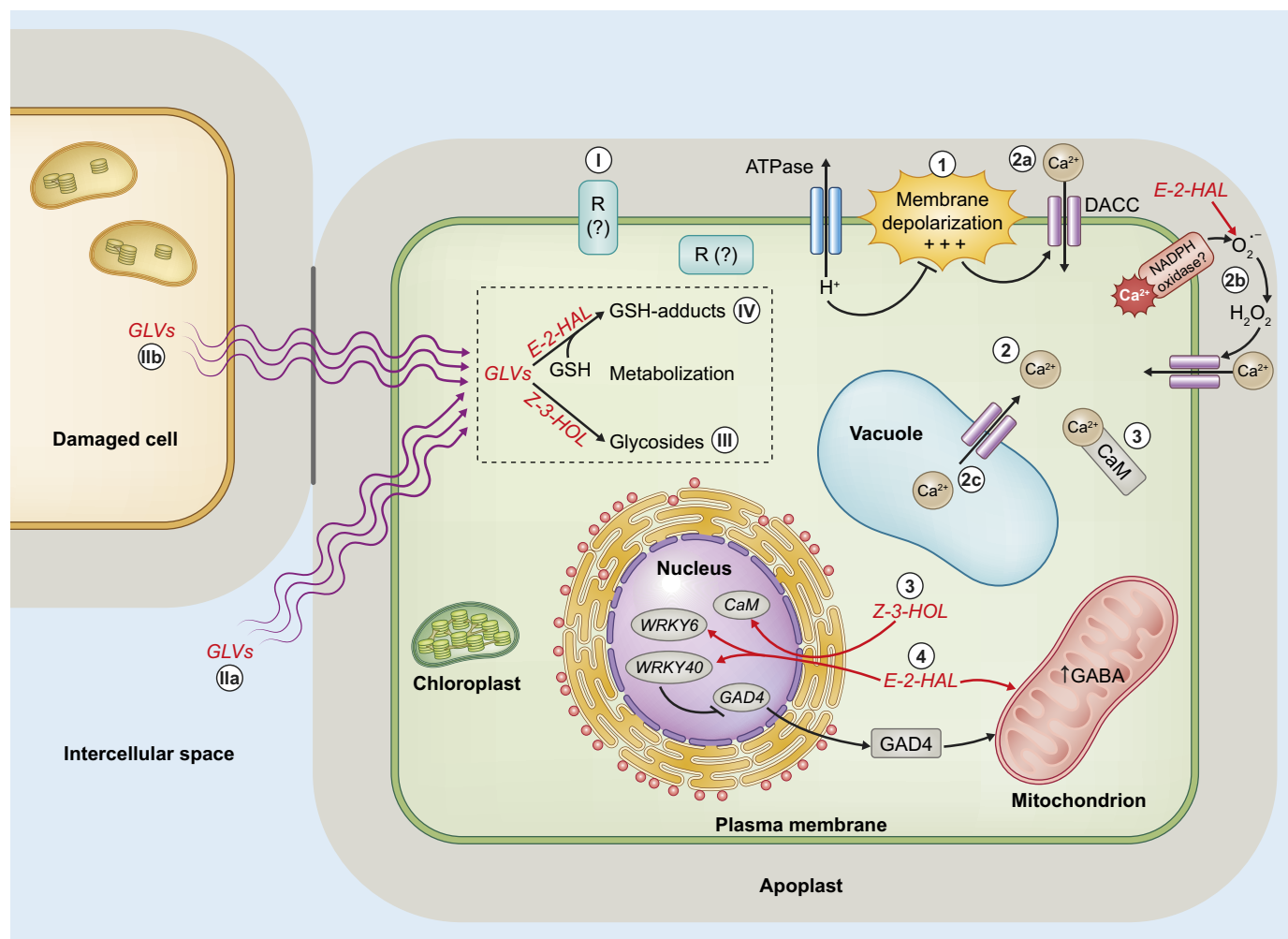


Fig. 6 Signal transduction by green leaf volatiles (GLVs). Early events in leaves upon GLV exposure include: (1) plasma membrane potential (V_m) depolarization (tomato), and (2) an increase in $[Ca^{2+}]_{cyt}$ (Arabidopsis and tomato). The increase in $[Ca^{2+}]_{cyt}$ can be caused by the activation of a depolarization-activated calcium channel (DACC) (2a), a reactive oxygen species (ROS)-activated calcium channel (2b) or via the release of Ca^{2+} from vacuoles (2c). Exposure of maize leaves to Z-3-hexenol (Z-3-HOL) increases the transcript abundance of calmodulin (CaM) (3). Other early signalling responses include the transcriptional activation of WRKY40 and WRKY6 by E-2-hexenal (E-2-HAL) in Arabidopsis (4). These transcription factors negatively regulate the transcription of glutamate decarboxylase 4 (GAD4). While it has been assumed that plants can recognize GLVs by specific protein receptors (R), they have not been identified yet (I). GLVs can at least partly enter the plant cell passively (II). These GLVs either originate from neighbouring leaves or plants and most likely enter the plant via the stomata (IIa), or are produced in damaged cells to reach intact neighbouring cells (IIb). GLVs that enter the cell can be metabolized in the cytoplasm by glycosylation (III) or glutathionylation (IV). The middle lamella is only indicated as a black line between the two plant cells. GSH, glutathione.

transcriptional activation of WRKY40 and 6 by E-2-HAL in Arabidopsis (Mirabella *et al.*, 2008, 2015). Early signalling responses include not only transcriptional, but also metabolic changes; an increased release of free FAs within 15 min of exposure to Z-3-HOL has recently been reported to be a common feature of several plant species, including monocots and dicots. In maize seedlings, this increase in free FAs seems to be involved in plant priming upon herbivory (Li *et al.*, 2016).

XI. GLVs influence the C/N metabolism

In order to mount their defences, plants need to redistribute energy and metabolites towards the defence machinery (Berger *et al.*, 2007; Bolton, 2009). Glutamate plays an important role in this (Seifi *et al.*, 2013): after pathogen attack, plants can maintain cell

viability by maintaining glutamate concentrations in infected tissue and replenishment of the tricarboxylic acid (TCA) cycle through the GABA shunt. This scenario is efficient when a plant is attacked by a necrotrophic pathogen. However, plants can also induce cell death by depleting glutamate concentrations and exhausting the TCA cycle. This strategy is used when a plant is under attack by a biotrophic pathogen. These strategies have been named endurance and evasion, respectively (Seifi *et al.*, 2013). The interference of pathogens with this C/N mechanism constitutes an important part of the infection strategy of several, often toxigenic, fungal pathogens such as *F. graminearum* (Audenaert *et al.*, 2013), *Alternaria* spp. (Klotz, 1988) and *Cochliobolus* spp. (Stergiopoulos *et al.*, 2013). Changes in the primary metabolism, including nitrogen assimilation, are also a well-known phenomenon upon insect herbivory (Zhou *et al.*, 2015); aphid attack induced the

expression of a nitrite reductase in sorghum (Zhu-Salzman *et al.*, 2004) and a glutamate synthase (GOGAT) in *N. attenuata* (Voelckel *et al.*, 2004). Both enzymes are required for the nitrogen assimilation into glutamate.

No link between C/N metabolism and GLVs was expected until recent work on *Arabidopsis* revealed up-regulation of glutamate decarboxylase 4 (*GAD4*), a key enzyme in the GABA shunt, after plants were exposed to E-2-HAL, coinciding with an increase in GABA concentrations (Mirabella *et al.*, 2008, 2015). Interestingly, the same study showed that E-2-HAL also induces expression of the transcription factor WRKY40, which negatively regulates *GAD4* expression. Microarray data from maize plants that had been exposed to Z-3-HOL for 60 min showed that transcript abundance of glutamine-fructose-6-phosphate transaminase 2, which converts glutamine to glutamate, was significantly up-regulated. This suggests that N is mobilized towards the exposed tissue under the form of glutamine, which is locally converted to glutamate (Engelberth *et al.*, 2013). Together, these data point to a model in which plant cells upon GLV exposure aim to maintain cell viability through TCA replenishment and nitrogen remobilization. The importance of the TCA replenishment in mounting a successful defence remains enigmatic.

XII. Interaction with plant hormones

Green leaf volatiles have mostly been studied in the context of plant–insect interactions, of which defence has generally been attributed to JA (Pieterse *et al.*, 2012). Consequently, most research on GLVs has focused on the effect of GLV on JA signalling. However, over the last decade it has become increasingly apparent that plant defence is regulated by a network of plant hormones and that a complex grid of crosstalk between these hormonal pathways moulds the outcome of the plant–insect/pathogen interaction (De Vleeschauwer *et al.*, 2014; Nguyen *et al.*, 2016).

The interaction between GLVs and JA has often been reported (Halitschke *et al.*, 2004). For example, treatment with GLVs resulted in increased transcript abundance of JA biosynthesis genes in *Arabidopsis* (Bate & Rothstein, 1998; Kishimoto *et al.*, 2005, 2006b), maize (Engelberth *et al.*, 2004), lima bean (Arimura *et al.*, 2001), citrus (Gomi *et al.*, 2003), and tea (Xin *et al.*, 2015), but not in tomato (Sugimoto *et al.*, 2014).

Furthermore, exogenously applied JA or GLVs induce Z-3-HOL, E-2-HAL and Z-3-HAC production in tomato plants and cereals (Wei *et al.*, 2011; Piesik *et al.*, 2013), creating a positive feedback loop. This mechanism warns plants of an impending fungus/insect attack, allowing them to induce or prime defences in systemic tissue as well as in neighbouring plants.

Wei *et al.* (2011) showed that *35S::prosys* tomato plants with constitutive JA signalling constantly released Z-3-HOL, even in the absence of damage by insect herbivores. They hypothesized that overexpressing the prosystemin gene enhances expression of *LOX* and *HPL*, which subsequently leads to higher GLV production. Tomato mutants defective in JA biosynthesis (*spr2*) did not abolish Z-3-HOL production after *S. exigua* damage, which suggests that GLV production is not exclusively governed by JA signalling. Bate & Rothstein (1998) found that treatment of *Arabidopsis* seedlings

with E-2-HAL induced expression of genes involved in the oxylipin biosynthesis and phenylpropanoid pathway. However, no induction of pathogenesis-related (PR) proteins was found. Interestingly, seed germination was reduced by E-2-HAL in the wild-type as well as in the JA-insensitive mutant (*jar1-1*), suggesting that E-2-HAL perception and signalling are independent of the JA pathway.

More recently, attention to the effect of GLVs on plant hormones has shifted from a focused view on JA biosynthesis and signalling to a more holistic approach using transcriptomics. Mirabella *et al.* (2015) found that gene activation by E-2-HAL in *Arabidopsis* was most closely related to gene activation after SA treatment (50%), followed by ABA (29%) and JA (13%), whereas ethylene treatment only showed a 1% overlap with E-2-HAL treatment. While these percentages give an idea of the overlap with other defence hormones, they do not give information on the signalling pathways that are activated. In other words, a 13% overlap may include all genes that play crucial roles in defence signalling, while the 50% might not include important defence genes. While -omics studies in general provide an unbiased look, further studies are needed to elucidate the underlying mechanism. The same study additionally reported that one-third of the up-regulated genes by E-2-HAL were unique for the E-2-HAL treatment and independent of aforementioned plant hormones. This confirmed earlier work from the same research group which exhibited an inhibitory effect of E-2-HAL on root growth of *Arabidopsis*, which was independent of JA, SA, ABA and ethylene signalling (Mirabella *et al.*, 2008). This partly contradicts the study by Kishimoto *et al.* (2006b) which reported that the induction of defence genes in *Arabidopsis*, after treatment with E-2-HAL and Z-3-HAL, was significantly repressed in JA and ethylene-deficient signalling mutants, but did not have an effect in the SA-insensitive mutant. As mentioned earlier, this raises the question of whether different GLVs induce different responses and whether knowledge gathered on *Arabidopsis* can be transferred to other crops.

In summary, GLVs have an inducing effect on JA biosynthesis and JA-dependent signalling. However, evidence provided by different studies suggests that GLV signalling is not exclusively governed by JA-dependent signalling, which might point to a separate, yet unknown, signalling pathway. To obtain a better insight into the role of GLVs in the plant hormone network, more research on the effect of GLVs on the plant hormone (signalling) network is needed.

XIII. General conclusions and unanswered questions

A meta-analysis approach offers a holistic view of GLV literature and allows us to uncover as yet unknown factors in the production of GLVs. However, the results must be carefully interpreted, and conclusions drawn from this analysis cannot serve as a 'one size fits all' for each plant–pathogen/herbivore interaction. Nonetheless, some surprising trends became apparent. The high induction of GLV production upon fungal infection, compared with herbivore infestation, remained until now largely overlooked in literature, but opens up an exciting field of research. Evidence is accumulating that herbivores modulate GLV emission. Besides suppression or enhancement of GLV production, herbivores can also increase

the conversion from Z-3- to E-2-GLV. Furthermore, there was a clear divergence between monocots and dicots which should be taken into account in further research and applications.

Despite the progress that has been made in recent years in unravelling the functions and mechanisms underlying GLV perception and signalling, a lot of questions remain unanswered, including key ones such as, how do plants perceive GLVs and via which mechanism is the signal transduced? Transcriptomic studies and forward genetics have been used to make the first steps towards understanding of the early signalling mechanisms in plants. With genes specifically induced by E-2-HAL available, upstream components can be identified through mutant screens. Such strategies, but also more chemically related approaches, are necessary to understand the mechanisms behind GLV perception in plants. Do generalist herbivores and necrotrophic fungi induce GLV emissions differently compared with specialist herbivores and biotrophic fungi? What is the underlying reason for the difference in GLV production between monocots and dicots? Comparative studies on specialist vs generalist herbivores, necrotrophic vs biotrophic fungi and monocots vs dicots will help us to understand how different plants react to different kinds of stress. Do GLVs interact solely with JA, or do other plant hormones also play a role? Plant hormone treatment experiments and use of knockout/overexpressing mutants in signalling pathways may aid in identifying synergistic or antagonistic interactions with GLV production and signalling. What is the ecological relevance of Z-3- to E-2-conversion of GLVs for plants and insects? The recent identification of a (3Z):(2E)-enal isomerase and the engineering and characterization of knockdown or knockout lines will help researchers to understand the physiological function of this enzyme in plants and insects. Can GLVs be used in agronomic practices, for example, as alternatives for pesticides? Long-term exposure experiments are, however, mandatory to determine whether (intermittent) GLV exposure entails fitness costs.

We strongly believe that unravelling these questions is of paramount importance and will provide a deeper understanding of a trait that has been preserved throughout the plant kingdom and serves as a general stress signal against (a)biotic stress.

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References

- Acevedo FE, Rivera-Vega LJ, Chung SH, Ray S, Felton GW. 2015. Cues from chewing insects – the intersection of DAMPs, HAMPs, MAMPs and effectors. *Current Opinion in Plant Biology* 26: 80–86.
- Allmann S, Baldwin IT. 2010. Insects betray themselves in nature to predators by rapid isomerization of green leaf volatiles. *Science* 329: 1075–1078.
- Allmann S, Halitschke R, Schuurink RC, Baldwin IT. 2010. Oxylipin channelling in *Nicotiana attenuata*: lipoxygenase 2 supplies substrates for green leaf volatile production. *Plant, Cell & Environment* 33: 2028–2040.
- Allmann S, Späthe A, Bisch-Knaden S, Kallenbach M, Reinecke A, Sachse S, Baldwin IT, Hansson BS. 2013. Feeding-induced rearrangement of green leaf volatiles reduces moth oviposition. *eLife* 2: e00421.
- Alm  ras E, Stolz S, Vollenweider S, Reymond P, M  ne-Saffran   L, Farmer EE. 2003. Reactive electrophile species activate defense gene expression in *Arabidopsis*. *Plant Journal* 34: 205–216.
- Amey M, Audenaert K, De Zutter N, Steppe K, Van Meulebroek L, Vanhaecke L, De Vleeschauwer D, Haesaert G, Smaghe G. 2015. Priming of wheat with the green leaf volatile Z-3-hexenyl acetate enhances defense against *Fusarium graminearum* but boosts deoxynivalenol production. *Plant Physiology* 167: 1671–1684.
- Andreou A, Feussner I. 2009. Lipoxygenases – structure and reaction mechanism. *Phytochemistry* 70: 1504–1510.
- Arimura G, Ozawa R, Horiuchi J, Nishioka T, Takabayashi J. 2001. Plant–plant interactions mediated by volatiles emitted from plants infested by spider mites. *Biochemical Systematics and Ecology* 29: 1049–1061.
- Asai N, Nishioka T, Takabayashi J, Furuichi T. 2009. Plant volatiles regulate the activities of Ca²⁺-permeable channels and promote cytoplasmic calcium transients in *Arabidopsis* leaf cells. *Plant Signaling & Behavior* 4: 294–300.
- Audenaert K, Vanheule A, H  fte M, Haesaert G. 2013. Deoxynivalenol: a major player in the multifaceted response of *Fusarium* to its environment. *Toxins* 6: 1–19.
- Baldwin IT, Schultz JC. 1983. Rapid changes in tree leaf chemistry induced by damage: evidence for communication between plants. *Science* 221: 277–279.
- Bate NJ, Rothstein SJ. 1998. C₆-volatiles derived from the lipoxygenase pathway induce a subset of defense-related genes. *Plant Journal* 16: 561–569.
- Berger S, Sinha AK, Roitsch T. 2007. Plant physiology meets phytopathology: plant primary metabolism and plant–pathogen interactions. *Journal of Experimental Botany* 58: 4019–4026.
- Bl  mke A, Falter C, Herrfurth C, Sode B, Bode R, Sch  fer W, Feussner I, Voigt CA. 2014. Secreted fungal effector lipase releases free fatty acids to inhibit innate immunity-related callose formation during wheat head infection. *Plant Physiology* 165: 346–358.
- Bolton MD. 2009. Primary metabolism and plant defense-fuel for the fire. *Molecular Plant–Microbe Interactions* 22: 487–497.
- Bonaventure G. 2014. Lipases and the biosynthesis of free oxylipins in plants. *Plant Signaling & Behavior* 9: e28429.
- Bonaventure G, Schuck S, Baldwin IT. 2011. Revealing complexity and specificity in the activation of lipase-mediated oxylipin biosynthesis: a specific role of the *Nicotiana attenuata* GLA1 lipase in the activation of jasmonic acid biosynthesis in leaves and roots. *Plant, Cell & Environment* 34: 1507–1520.
- Brand A, Gow NA. 2009. Mechanisms of hypha orientation of fungi. *Current Opinion in Microbiology* 12: 350–357.
- Bruce TJA, Pickett JA. 2011. Perception of plant volatile blends by herbivorous insects – finding the right mix. *Phytochemistry* 72: 1605–1611.
- Bruinsma M, Posthumus MA, Mumm R, Mueller MJ, van Loon JJ, Dicke M. 2009. Jasmonic acid-induced volatiles of *Brassica oleracea* attract parasitoids: effects of time and dose, and comparison with induction by herbivores. *Journal of Experimental Botany* 60: 2575–2587.
- Carey AF, Carlson JR. 2011. Insect olfaction from model systems to disease control. *Proceedings of the National Academy of Sciences, USA* 108: 12987–12995.
- Chehab EW, Kaspi R, Savchenko T, Rowe H, Negre-Zakharov F, Kliebenstein D, Dehesh K. 2008. Distinct roles of jasmonates and aldehydes in plant-defense responses. *PLoS ONE* 3: e1904.
- Christensen SA, Nemchenko A, Borrego E, Murray I, Sobhy IS, Bosak L, DeBlasio S, Erb M, Robert CAM, Vaughn KA *et al.* 2013. The maize lipoxygenase, ZmLOX10, mediates green leaf volatile, jasmonate and herbivore-induced plant volatile production for defense against insect attack. *Plant Journal* 74: 59–73.
- Clavijo McCormick A, Unsicker SB, Gershenzon J. 2012. The specificity of herbivore-induced plant volatiles in attracting herbivore enemies. *Trends in Plant Science* 17: 303–310.
- Copolovici L, K  nnaste A, Pazouki L, Niinemets   . 2012. Emissions of green leaf volatiles and terpenoids from *Solanum lycopersicum* are quantitatively related to the severity of cold and heat shock treatments. *Journal of Plant Physiology* 169: 664–672.
- Curtius T, Franzen H. 1914.   ber die chemischen Bestandteile gr  ner Pflanzen.   ber die fl  chtigen Bestandteile der Hainbuchenbl  tter. *European Journal of Organic Chemistry* 404: 93–130.
- D'Auria JC, Pichersky E, Schaub A, Hansel A, Gershenzon J. 2007. Characterization of a BAHD acyltransferase responsible for producing the green

- leaf volatile (Z)-3-hexen-1-yl acetate in *Arabidopsis thaliana*. *Plant Journal* 49: 194–207.
- Davoine C, Falletti O, Douki T, Iacazio G, Ennar N, Montillet J-L, Triantaphyllides C. 2006. Adducts of oxylipin electrophiles to glutathione reflect a 13 specificity of the downstream lipoxygenase pathway in the tobacco hypersensitive response. *Plant Physiology* 140: 1484–1493.
- De Lucca AJ, Carter-Wientjes CH, Boue S, Bhatnagar D. 2011. Volatile *trans*-2-hexenal, a soybean aldehyde, inhibits *Aspergillus flavus* growth and aflatoxin production in corn. *Journal of Food Science* 76: M381–M386.
- De Moraes CM, Mescher MC, Tumlinson JH. 2001. Caterpillar-induced nocturnal plant volatiles repel conspecific females. *Nature* 410: 577–580.
- De Vleeschauwer D, Xu J, Höfte M. 2014. Making sense of hormone-mediated defense networking: from rice to *Arabidopsis*. *Frontiers in Plant Science* 5: 611.
- De Zutter N, Audenaert K, Ameye M, De Boevre M, De Saeger S, Haesaert G, Smaghe G. 2017. The plant response induced in wheat ears by a combined attack of *Sitobion avenae* aphids and *Fusarium graminearum* boosts fungal infection and deoxynivalenol production. *Molecular Plant Pathology* 18: 98–109.
- Dicke M, Baldwin IT. 2009. The evolutionary context for herbivore-induced plant volatiles: beyond the 'cry for help'. *Trends in Plant Science* 15: 167–175.
- Ding L, Xu H, Yi H, Yang L, Kong Z, Zhang L, Xue S, Jia H, Ma Z. 2011. Resistance to hemi-biotrophic *F. graminearum* infection is associated with coordinated and ordered expression of diverse defense signaling pathways. *PLoS ONE* 6: e19008.
- Drakulic J, Ajigboye O, Swarup R, Bruce T, Ray RV. 2016. Aphid infestation increases *Fusarium langsethiae* and T-2 and HT-2 mycotoxins in wheat. *Applied and Environmental Microbiology* 82: 6548–6556.
- Drakulic J, Caulfield J, Woodcock C, Jones SP, Linforth R, Bruce TJ, Ray RV. 2015. Sharing a host plant (Wheat [*Triticum aestivum*]) increases the fitness of *Fusarium graminearum* and the severity of *Fusarium* head blight but reduces the fitness of grain aphids (*Sitobion avenae*). *Applied and Environmental Microbiology* 81: 3492–3501.
- Dudareva N, Negre F, Nagegowda DA, Orlova I. 2006. Plant volatiles: recent advances and future perspectives. *Critical Reviews in Plant Sciences* 25: 417–440.
- Duran-Flores D, Heil M. 2016. Sources of specificity in plant damaged-self recognition. *Current Opinion in Plant Biology* 32: 77–87.
- El Oirdi M, El Rahman TA, Rigano L, El Hadrami A, Rodriguez MC, Daayf F, Vojnov A, Bouarab K. 2011. *Botrytis cinerea* manipulates the antagonistic effects between immune pathways to promote disease development in tomato. *Plant Cell* 23: 2405–2421.
- Ellinger D, Kubigsteltig II. 2010. Involvement of DAD1-like lipases in response to salt and osmotic stress in *Arabidopsis thaliana*. *Plant Signaling & Behavior* 5: 1269–1271.
- Ellinger D, Stingl N, Kubigsteltig II, Bals T, Juenger M, Pollmann S, Berger S, Schuenemann D, Mueller MJ. 2010. DONGLE and DEFECTIVE IN ANTHOR DEHISCENCE1 lipases are not essential for wound- and pathogen-induced jasmonate biosynthesis: redundant lipases contribute to jasmonate formation. *Plant Physiology* 153: 114–127.
- Engelberth J, Alborn HT, Schmelz EA, Tumlinson JH. 2004. Airborne signals prime plants against insect herbivore attack. *Proceedings of the National Academy of Sciences, USA* 101: 1781–1785.
- Engelberth J, Contreras CF, Dalvi C, Li T, Engelberth M. 2013. Early transcriptome analyses of Z-3-hexenol-treated *Zea mays* revealed distinct transcriptional networks and anti-herbivore defense potential of green leaf volatiles. *PLoS ONE* 8: e77465.
- Engelberth J, Seidl-Adams I, Schultz JC, Tumlinson JH. 2007. Insect elicitors and exposure to green leaf volatiles differentially upregulate major octadecanoids and transcripts of 12-oxo phytodienoic acid reductases in *Zea mays*. *Molecular Plant-Microbe Interactions* 20: 707–716.
- Fall R, Karl T, Hansel A, Jordan A, Lindinger W. 1999. Volatile organic compounds emitted after leaf wounding: on-line analysis by proton-transfer-reaction mass spectrometry. *Journal of Geophysical Research: Atmospheres* 104: 15963–15974.
- Farag M, Fokar M, Abd H, Zhang H, Allen R, Paré P. 2005. (Z)-3-hexenol induces defense genes and downstream metabolites in maize. *Planta* 220: 900–909.
- Farag MA, Paré PW. 2002. C₆-Green leaf volatiles trigger local and systemic VOC emissions in tomato. *Phytochemistry* 61: 545–554.
- Farmaki T, Sanmartín M, Jiménez P, Paneque M, Sanz C, Vancanneyt G, León J, Sánchez-Serrano JJ. 2007. Differential distribution of the lipoxygenase pathway enzymes within potato chloroplasts. *Journal of Experimental Botany* 58: 555–568.
- Farmer EE, Davoine C. 2007. Reactive electrophile species. *Current Opinion in Plant Biology* 10: 380–386.
- Frøehlich JE, Itoh A, Howe GA. 2001. Tomato allene oxide synthase and fatty acid hydroperoxide lyase, two cytochrome P450s involved in oxylipin metabolism, are targeted to different membranes of chloroplast envelope. *Plant Physiology* 125: 306–317.
- Frost CJ, Mescher MC, Dervinis C, Davis JM, Carlson JE, De Moraes CM. 2008. Priming defense genes and metabolites in hybrid poplar by the green leaf volatile *cis*-3-hexenyl acetate. *New Phytologist* 180: 722–733.
- Gadenne C, Barrozo RB, Anton S. 2016. Plasticity in insect olfaction: to smell or not to smell? *Annual Review of Entomology* 61: 317–333.
- Gaquerel E, Weinhold A, Baldwin IT. 2009. Molecular interactions between the specialist herbivore *Manduca sexta* (Lepidoptera, Sphingidae) and its natural host *Nicotiana attenuata*. VIII. An unbiased GCxGC-ToFMS analysis of the plant's elicited volatile emissions. *Plant Physiology* 149: 1408–1423.
- Gobel C, Feussner I. 2009. Methods for the analysis of oxylipins in plants. *Phytochemistry* 70: 1485–1503.
- Gomi K, Yamasaki Y, Yamamoto H, Akimitsu K. 2003. Characterization of a hydroperoxide lyase gene and effect of C₆-volatiles on expression of genes of the oxylipin metabolism in Citrus. *Journal of Plant Physiology* 160: 1219–1231.
- Halitschke R, Baldwin IT. 2003. Antisense LOX expression increases herbivore performance by decreasing defense responses and inhibiting growth-related transcriptional reorganization in *Nicotiana attenuata*. *Plant Journal* 36: 794–807.
- Halitschke R, Stenberg JA, Kessler D, Kessler A, Baldwin IT. 2008. Shared signals – 'alarm calls' from plants increase apparency to herbivores and their enemies in nature. *Ecology Letters* 11: 24–34.
- Halitschke R, Ziegler J, Keinänen M, Baldwin IT. 2004. Silencing of hydroperoxide lyase and allene oxide synthase reveals substrate and defense signaling crosstalk in *Nicotiana attenuata*. *Plant Journal* 40: 35–46.
- Hamilton-Kemp TR, McCracken CT, Loughrin JH, Andersen RA, Hildebrand DF. 1992. Effects of some natural volatile compounds on the pathogenic fungi *Alternaria alternata* and *Botrytis cinerea*. *Journal of Chemical Ecology* 18: 1083–1091.
- Han B, Zhang Q-H, Byers JA. 2012. Attraction of the tea aphid, *Toxoptera aurantii*, to combinations of volatiles and colors related to tea plants. *Entomologia Experimentalis Et Applicata* 144: 258–269.
- Hansson BS, Larsson MC, Leal WS. 1999. Green leaf volatile-detecting olfactory receptor neurones display very high sensitivity and specificity in a scarab beetle. *Physiological Entomology* 24: 121–126.
- ul Hassan MN, Zainal Z, Ismail I. 2015. Green leaf volatiles: biosynthesis, biological functions and their applications in biotechnology. *Plant biotechnology Journal* 13: 727–739.
- Hatanaka A. 1993. The biogenesis of green odour by green leaves. *Phytochemistry* 34: 1201–1218.
- Hatanaka A, Kajiwaru T, Sekiya J. 1987. Biosynthetic pathway for C₆-aldehydes formation from linolenic acid in green leaves. *Chemistry and Physics of Lipids* 44: 341–361.
- Hedges LV. 1981. Distribution theory for Glass's estimator of effect size and related estimators. *Journal of Educational and Behavioral Statistics* 6: 107–128.
- Heil M. 2014. Herbivore-induced plant volatiles: targets, perception and unanswered questions. *New Phytologist* 204: 297–306.
- Heil M, Lion U, Boland W. 2008. Defense-inducing volatiles: in search of the active motif. *Journal of Chemical Ecology* 34: 601–604.
- Heil M, Ton J. 2008. Long-distance signalling in plant defence. *Trends in Plant Science* 13: 264–272.
- Hughes RK, De Domenico S, Santino A. 2009. Plant cytochrome CYP74 family: biochemical features, endocellular localisation, activation mechanism in plant defence and improvements for industrial applications. *ChemBioChem* 10: 1122–1133.
- Kant MR, Bleeker PM, Wijk MV, Schuurink RC, Haring MA. 2009. Chapter 14. Plant volatiles in defence. In: Loon LCV, ed. *Advances in botanical research*. Amsterdam, the Netherlands: Academic Press, 613–666.
- Kant MR, Jonckheere W, Knecht B, Lemos F, Liu J, Schimmel BCJ, Villarroel CA, Ataide LMS, Dermauw W, Glas JJ *et al.* 2015. Mechanisms and ecological

- consequences of plant defence induction and suppression in herbivore communities. *Annals of Botany* 115: 1015–1051.
- Karban R, Yang LH, Edwards KF. 2014. Volatile communication between plants that affects herbivory: a meta-analysis. *Ecology Letters* 17: 44–52.
- Kessler A, Baldwin IT. 2001. Defensive function of herbivore-induced plant volatile emissions in nature. *Science* 291: 2141–2144.
- Kessler A, Halitschke R, Diezel C, Baldwin IT. 2006. Priming of plant defense responses in nature by airborne signaling between *Artemisia tridentata* and *Nicotiana attenuata*. *Oecologia* 148: 280–292.
- Kishimoto K, Matsui K, Ozawa R, Takabayashi J. 2005. Volatile C6-aldehydes and allo-ocimene activate defense genes and induce resistance against *Botrytis cinerea* in *Arabidopsis thaliana*. *Plant and Cell Physiology* 46: 1093–1102.
- Kishimoto K, Matsui K, Ozawa R, Takabayashi J. 2006a. Components of C6-aldehyde-induced resistance in *Arabidopsis thaliana* against a necrotrophic fungal pathogen, *Botrytis cinerea*. *Plant Science* 170: 715–723.
- Kishimoto K, Matsui K, Ozawa R, Takabayashi J. 2006b. ETR1-, JAR1- and PAD2-dependent signaling pathways are involved in C6-aldehyde-induced defense responses of *Arabidopsis*. *Plant Science* 171: 415–423.
- Kishimoto K, Matsui K, Ozawa R, Takabayashi J. 2008. Direct fungicidal activities of C6-aldehydes are important constituents for defense responses in *Arabidopsis* against *Botrytis cinerea*. *Phytochemistry* 69: 2127–2132.
- Klotz MG. 1988. The action of tentoxin on membrane processes in plants. *Physiologia Plantarum* 74: 575–582.
- Kluth S, Kruess A, Tscharnkte T. 2002. Insects as vectors of plant pathogens: mutualistic and antagonistic interactions. *Oecologia* 133: 193–199.
- Kost C, Heil M. 2006. Herbivore-induced plant volatiles induce an indirect defence in neighbouring plants. *Journal of Ecology* 94: 619–628.
- Kunishima M, Yamauchi Y, Mizutani M, Kuse M, Takikawa H, Sugimoto Y. 2016. Identification of (Z)-3-(E)-2-hexenal isomerases essential to the production of the leaf aldehyde in plants. *Journal of Biological Chemistry* 291(27): 14023–14033.
- Leal WS. 2013. Odorant reception in insects: roles of receptors, binding proteins, and degrading enzymes. *Annual Review of Entomology* 58: 373–391.
- Li T, Cofer T, Engelberth M, Engelberth J. 2016. Defense priming and jasmonates: a role for free fatty acids in insect elicitor-induced long distance signaling. *Plants* 5: 5.
- Liavonchanka A, Feussner N. 2006. Lipoxygenases: occurrence, functions and catalysis. *Journal of Plant Physiology* 163: 348–357.
- Liebler DC. 2008. Protein damage by reactive electrophiles: targets and consequences. *Chemical Research in Toxicology* 21: 117–128.
- Liu X, Li F, Tang J, Wang W, Zhang F, Wang G, Chu J, Yan C, Wang T, Chu C *et al.* 2012. Activation of the jasmonic acid pathway by depletion of the hydroperoxide lyase OsHPL3 reveals crosstalk between the HPL and AOS branches of the oxylipin pathway in rice. *PLoS ONE* 7: e50089.
- Loreto F, Barta C, Brilli F, Nogueis I. 2006. On the induction of volatile organic compound emissions by plants as consequence of wounding or fluctuations of light and temperature. *Plant, Cell & Environment* 29: 1820–1828.
- Maja MM, Kasurinen A, Yli-Pirilä P, Joutsensaari J, Klemola T, Holopainen T, Holopainen JK. 2014. Contrasting responses of silver birch VOC emissions to short- and long-term herbivory. *Tree Physiology* 34: 241–252.
- Maor R, Shirasu K. 2005. The arms race continues: battle strategies between plants and fungal pathogens. *Current Opinion in Microbiology* 8: 399–404.
- Martinez-Medina A, Flors V, Heil M, Mauch-Mani B, Pieterse CM, Pozo MJ, Ton J, van Dam NM, Conrath U. 2016. Recognizing plant defense priming. *Trends in Plant Science* 21: 818–822.
- Matsui K. 2006. Green leaf volatiles: hydroperoxide lyase pathway of oxylipin metabolism. *Current Opinion in Plant Biology* 9: 274–280.
- Matsui K. 2016. A portion of plant airborne communication is endorsed by uptake and metabolism of volatile organic compounds. *Current Opinion in Plant Biology* 32: 24–30.
- Matsui K, Kurishita S, Hisamitsu A, Kajiwaru T. 2000. A lipid-hydrolysing activity involved in hexenal formation. *Biochemical Society Transactions* 28: 857–860.
- Matsui K, Sugimoto K, Mano J, Ozawa R, Takabayashi J. 2012. Differential metabolisms of green leaf volatiles in injured and intact parts of a wounded leaf meet distinct ecophysiological requirements. *PLoS ONE* 7: e36433.
- Mattiacci L, Dicke M, Posthumus MA. 1995. beta-Glucosidase – an elicitor of herbivore-induced plant odor that attracts host-searching parasitic wasps. *Proceedings of the National Academy of Sciences, USA* 92: 2036–2040.
- Mirabella R, Rauwerda H, Allmann S, Scala A, Spyropoulou EA, de Vries M, Boersma MR, Breit TM, Haring MA, Schuurink RC. 2015. WRKY40 and WRKY6 act downstream of the green leaf volatile E-2-hexenal in *Arabidopsis*. *Plant Journal* 83: 1082–1096.
- Mirabella R, Rauwerda H, Struys EA, Jakobs C, Triantaphylides C, Haring MA, Schuurink RC. 2008. The *Arabidopsis* *her1* mutant implicates GABA in E-2-hexenal responsiveness. *Plant Journal* 53: 197–213.
- Mithöfer A, Wanner G, Boland W. 2005. Effects of feeding *Spodoptera littoralis* on lima bean leaves. II. Continuous mechanical wounding resembling insect feeding is sufficient to elicit herbivory-related volatile emission. *Plant Physiology* 137: 1160–1168.
- Möbus N, Hertweck C. 2009. Fungal phytotoxins as mediators of virulence. *Current Opinion in Plant Biology* 12: 390–398.
- Mochizuki S, Sugimoto K, Koeduka T, Matsui K. 2016. *Arabidopsis* lipoxygenase 2 is essential for formation of green leaf volatiles and five-carbon volatiles. *FEBS Letters* 590: 1017–1027.
- Mueller MJ, Berger S. 2009. Reactive electrophilic oxylipins: pattern recognition and signalling. *Phytochemistry* 70: 1511–1521.
- Mwenda CM, Matsui K. 2014. The importance of lipoxygenase control in the production of green leaf volatiles by lipase-dependent and independent pathways. *Plant Biotechnology* 31: 445–452.
- Mwenda CM, Matsui K, Nishimura K, Koeduka T, Matsui K. 2015. Spatial expression of the *Arabidopsis* *hydroperoxide lyase* gene is controlled differently from that of the *allene oxide synthase* gene. *Journal of Plant Interactions* 10: 1–10.
- Nakamura S, Hatanaka A. 2002. Green-leaf-derived C6-aroma compounds with potent antibacterial action that act on both gram-negative and gram-positive bacteria. *Journal of Agricultural and Food Chemistry* 50: 7639–7644.
- Nakashima A, von Reuss SH, Tasaka H, Nomura M, Mochizuki S, Iijima Y, Aoki K, Shibata D, Boland W, Takabayashi J. 2013. Traumatins and dinortraumatins containing galactolipids in *Arabidopsis* their formation in tissue-disrupted leaves as counterparts of green leaf volatiles. *Journal of Biological Chemistry* 288: 26078–26088.
- Nguyen T, Nioi P, Pickett CB. 2009. The Nrf2-antioxidant response element signaling pathway and its activation by oxidative stress. *Journal of Biological Chemistry* 284: 13291–13295.
- Nguyen D, Rieu I, Mariani C, van Dam NM. 2016. How plants handle multiple stresses: hormonal interactions underlying responses to abiotic stress and insect herbivory. *Plant Molecular Biology* 91: 1–14.
- Noordermeer MA, Veldink GA, Vliegthart JFG. 1999. Alfalfa contains substantial 9-hydroperoxide lyase activity and a 3Z:2E-enal isomerase. *FEBS Letters* 443: 201–204.
- Noordermeer MA, Veldink GA, Vliegthart JF. 2001. Fatty acid hydroperoxide lyase: a plant cytochrome P450 enzyme involved in wound healing and pest resistance. *ChemBioChem* 2: 494–504.
- Obara N, Hasegawa M, Kodama O. 2002. Induced volatiles in elicitor-treated and rice blast fungus-inoculated rice leaves. *Bioscience Biotechnology and Biochemistry* 66: 2549–2559.
- Phillips DR, Matthew JA, Reynolds J, Fenwick GR. 1979. Partial purification and properties of a cis-3: trans-2-enal isomerase from cucumber fruit. *Phytochemistry* 18: 401–404.
- Piesik D, Panka D, Delaney KJ, Skoczek A, Lamparski R, Weaver DK. 2011. Cereal crop volatile organic compound induction after mechanical injury, beetle herbivory (*Oulema* spp.), or fungal infection (*Fusarium* spp.). *Journal of Plant Physiology* 168: 878–886.
- Piesik D, Panka D, Jeske M, Wenda-Piesik A, Delaney KJ, Weaver DK. 2013. Volatile induction of infected and neighbouring uninfected plants potentially influence attraction/repellence of a cereal herbivore. *Journal of Applied Entomology* 137: 296–309.
- Pieterse CMJ, Van der Does D, Zamioudis C, Leon-Reyes A, Van Wees SCM. 2012. Hormonal modulation of plant immunity. *Annual Review of Cell and Developmental Biology* 28: 489–521.
- Ponzo C, Gols R, Pieterse CMJ, Dicke M. 2013. Ecological and phytohormonal aspects of plant volatile emission in response to single and dual infestations with herbivores and phytopathogens. *Functional Ecology* 27: 587–598.
- Prost I, Dhondt S, Rothe G, Vicente J, Rodriguez MJ, Kift N, Carbonne F, Griffiths G, Esquerre-Tugay M-T, Rosahl S *et al.* 2005. Evaluation of the

- antimicrobial activities of plant oxylipins supports their involvement in defense against pathogens. *Plant Physiology* 139: 1902–1913.
- Qi J, Zhou G, Yang L, Erb M, Lu Y, Sun X, Cheng J, Lou Y. 2011. The chloroplast-localized phospholipases D $\alpha 4$ and $\alpha 5$ regulate herbivore-induced direct and indirect defenses in rice. *Plant Physiology* 157: 1987–1999.
- Reinecke A, Ruther J, Hilker M. 2005. Electrophysiological and behavioural responses of *Melolontha melolontha* to saturated and unsaturated aliphatic alcohols. *Entomologia Experimentalis et Applicata* 115: 33–40.
- Rhoades DF. 1983. Responses of alder and willow to attack by tent caterpillars and webworms: evidence for pheromonal sensitivity of willows. In: Hedin PA, ed. *Plant resistance to insects*. Washington, DC, USA: American Chemical Society, 55–68.
- Röstlién T, Strandén M, Borg-Karlson AK, Mustaparta H. 2005. Olfactory receptor neurons in two heliothine moth species responding selectively to aliphatic green leaf volatiles, aromatic compounds, monoterpenes and sesquiterpenes of plant origin. *Chemical Senses* 30: 443–461.
- Rowen E, Kaplan I. 2016. Eco-evolutionary factors drive induced plant volatiles: a meta-analysis. *New Phytologist* 210: 284–294.
- Ruther J, Fürstenau B. 2005. Emission of herbivore-induced volatiles in absence of a herbivore-response of *Zea mays* to green leaf volatiles and terpenoids. *Zeitschrift für Naturforschung C* 60: 743–756.
- Savchenko T, Pearce IS, Ignatia L, Karban R, Dehesh K. 2012. Insect herbivores selectively suppress the HPL branch of the oxylipin pathway in host plants. *Plant Journal* 73: 653–662.
- Savchenko TV, Zastrijnaja OM, Klimov VV. 2014. Oxylipins and plant abiotic stress resistance. *Biochemistry* 79: 362–375.
- Scala A, Allmann S, Mirabella R, Haring MA, Schuurink RC. 2013. Green leaf volatiles: a plant's multifunctional weapon against herbivores and pathogens. *International Journal of Molecular Sciences* 14: 17781–17811.
- Schaller GE, Bleeker AB. 1995. Ethylene-binding sites generated in yeast expressing the *Arabidopsis ETR1* gene. *Science* 270: 1809–1811.
- Schmelz EA, Engelberth J, Alborn HT, Tumlinson JH III, Teal PEA. 2009. Phytohormone-based activity mapping of insect herbivore-produced elicitors. *Proceedings of the National Academy of Sciences, USA* 106: 653–657.
- Schwartzberg EG, Böröczky K, Tumlinson JH. 2011. Pea aphids, *Acyrtosiphon pisum*, suppress induced plant volatiles in broad bean, *Vicia faba*. *Journal of Chemical Ecology* 37: 1055–1062.
- Seifi HS, Van Bockhaven J, Angenon G, Höfte M. 2013. Glutamate metabolism in plant disease and defense: friend or foe? *Molecular Plant–Microbe Interactions* 26: 475–485.
- Shimoda T. 2010. A key volatile infochemical that elicits a strong olfactory response of the predatory mite *Neoseiulus californicus*, an important natural enemy of the two-spotted spider mite *Tetranychus urticae*. *Experimental and Applied Acarology* 50: 9–22.
- Shiojiri K, Kishimoto K, Ozawa R, Kugimiya S, Urashimo S, Arimura G, Horiuchi J, Nishioka T, Matsui K, Takabayashi J. 2006a. Changing green leaf volatile biosynthesis in plants: an approach for improving plant resistance against both herbivores and pathogens. *Proceedings of the National Academy of Sciences, USA* 103: 16672–16676.
- Shiojiri K, Ozawa R, Matsui K, Kishimoto K, Kugimiya S, Takabayashi J. 2006b. Role of the lipoxygenase/lyase pathway of host-foed plants in the host searching behavior of two parasitoid species, *Cotesia glomerata* and *Cotesia plutellae*. *Journal of Chemical Ecology* 32: 969–979.
- Stergiopoulos I, Collemare J, Mehrabi R, De Wit PJ. 2013. Phytotoxic secondary metabolites and peptides produced by plant pathogenic Dothideomycete fungi. *FEMS Microbiology Reviews* 37: 67–93.
- Sufang Z, Jianing W, Zhen Z, Le K. 2013. Rhythms of volatiles release from healthy and insect-damaged *Phaseolus vulgaris*. *Plant Signaling & Behavior* 8: e25759.
- Sugimoto K, Matsui K, Iijima Y, Akakabe Y, Muramoto S, Ozawa R, Uefune M, Sasaki R, Alamgir KM, Akitake S et al. 2014. Intake and transformation to a glycoside of (Z)-3-hexenol from infested neighbors reveals a mode of plant odor reception and defense. *Proceedings of the National Academy of Sciences, USA* 111: 7144–7149.
- Swarbreck SM, Colaço R, Davies JM. 2013. Plant calcium-permeable channels. *Plant Physiology* 163: 514–522.
- Sykietis GP, Bohmann D. 2008. Keap1/Nrf2 signaling regulates oxidative stress tolerance and lifespan in *Drosophila*. *Developmental Cell* 14: 76–85.
- Sykietis GP, Bohmann D. 2010. Stress-activated cap'n'collar transcription factors in aging and human disease. *Science Signaling* 3: re3.
- Tajul MI, Motoyama T, Hatanaka A, Sariah M, Osada H. 2012. Green-odour compounds have antifungal activity against the rice blast fungus *Magnaporthe oryzae*. *European Journal of Plant Pathology* 132: 91–100.
- Takamura H, Gardner HW. 1996. Oxygenation of (3Z)-alkenal to (2E)-4-hydroxy-2-alkenal in soybean seed (*Glycine max* L.). *Biochimica et Biophysica Acta–Lipids and Lipid Metabolism* 1303: 83–91.
- Tong X, Qi J, Zhu X, Mao B, Zeng L, Wang B, Li Q, Zhou G, Xu X, Lou Y. 2012. The rice hydroperoxide lyase OsHPL3 functions in defense responses by modulating the oxylipin pathway. *Plant Journal* 71: 763–775.
- Trikalinos TA, Salanti G, Zintzaras E, Ioannidis JP. 2008. Meta-analysis methods. *Advances in Genetics* 60: 311–334.
- Turlings TC, Lengwiler UB, Bernasconi ML, Wechsler D. 1998. Timing of induced volatile emissions in maize seedlings. *Planta* 207: 146–152.
- Turlings TC, Tumlinson JH, Heath RR, Proveaux AT, Doolittle RE. 1991. Isolation and identification of allelochemicals that attract the larval parasitoid, *Cotesia marginiventris* (Cresson), to the microhabitat of one of its hosts. *Journal of Chemical Ecology* 17: 2235–2251.
- Turlings TCJ, Wäckers F. 2004. Recruitment of predators and parasitoids by herbivore-injured plants. In: Cardé RT, Millar JG, eds. *Advances in insect chemical ecology*. Cambridge, UK: Cambridge University Press, 21–75.
- Turrà D, El Ghalid M, Rossi F, Di Pietro A. 2015. Fungal pathogen uses sex pheromone receptor for chemotropic sensing of host plant signals. *Nature* 527: 521–524.
- Turrà D, Nordzieke D, Vitale S, El Ghalid M, Di Pietro A. 2016. Hyphal chemotropism in fungal pathogenicity. *Seminars in Cell & Developmental Biology* 57: 69–75.
- Verhage A, van Wees SC, Pieterse CM. 2010. Plant immunity: it's the hormones talking, but what do they say? *Plant Physiology* 154: 536–540.
- Visser JH, van Straten S, Maarse H. 1979. Isolation and identification of volatiles in the foliage of potato, *Solanum tuberosum*, a host plant of the colorado beetle, *Leptinotarsa decemlineata*. *Journal of Chemical Ecology* 5: 13–25.
- Voelckel C, Weisser WW, Baldwin IT. 2004. An analysis of plant–aphid interactions by different microarray hybridization strategies. *Molecular Ecology* 13: 3187–3195.
- Walton JD. 1996. Host-selective toxins: agents of compatibility. *Plant Cell* 8: 1723.
- Wang R, Shen W, Liu L, Jiang L, Liu Y, Su N, Wan J. 2008. A novel lipoxygenase gene from developing rice seeds confers dual position specificity and responds to wounding and insect attack. *Plant Molecular Biology* 66: 401–414.
- Webster B, Cardé RT. 2017. Use of habitat odour by host-seeking insects. *Biological Reviews* 92: 1241–1249.
- Wei J, Wang L, Zhao J, Li C, Ge F, Kang L. 2011. Ecological trade-offs between jasmonic acid-dependent direct and indirect plant defences in tritrophic interactions. *New Phytologist* 189: 557–567.
- Wenda-Piesik A. 2011. Volatile organic compound emissions by winter wheat plants (*Triticum aestivum* L.) under *Fusarium* spp. infestation and various abiotic conditions. *Polish Journal of Environmental Studies* 20: 1335–1342.
- Wenda-Piesik A, Piesik D, Ligor T, Buszewski B. 2010. Volatile organic compounds (VOCs) from cereal plants infested with crown rot: their identity and their capacity for inducing production of VOCs in uninfested plants. *International Journal of Pest Management* 56: 377–383.
- Xin Z, Li X, Li J, Chen Z, Sun X. 2015. Application of chemical elicitor (Z)-3-hexenol enhances direct and indirect plant defenses against tea geometrid *Ectropis obliqua*. *BioControl* 61: 1–12.
- Xin Z, Zhang L, Zhang Z, Chen Z, Sun X. 2014. A tea hydroperoxide lyase gene, *CsHPL1*, regulates tomato defense response against *Prodenia litura* (Fabricius) and *Alternaria alternata* f. sp. *lycopersici* by modulating green leaf volatiles (GLVs) release and jasmonic acid (JA) gene expression. *Plant Molecular Biology Reporter* 32: 62–69.
- Yamauchi Y, Kunishima M, Mizutani M, Sugimoto Y. 2015. Reactive short-chain leaf volatiles act as powerful inducers of abiotic stress-related gene expression. *Scientific Reports* 5: 8030.
- Yan C, Xie D. 2015. Jasmonate in plant defence: sentinel or double agent? *Plant Biotechnology Journal* 13: 1233–1240.
- Yan Z-G, Wang C-Z. 2006. Similar attractiveness of maize volatiles induced by *Helicoverpa armigera* and *Pseudaletia separata* to the generalist

- parasitoid *Camponotus chlorideae*. *Entomologia Experimentalis et Applicata* 118: 87–96.
- Zarate SI, Kempema LA, Walling LL. 2007. Silverleaf whitefly induces salicylic acid defenses and suppresses effectual jasmonic acid defenses. *Plant Physiology* 143: 866–875.
- Zebelo SA, Matsui K, Ozawa R, Maffei ME. 2012. Plasma membrane potential depolarization and cytosolic calcium flux are early events involved in tomato (*Solanum lycopersicon*) plant-to-plant communication. *Plant Science* 196: 93–100.
- Zeringue HJ Jr. 1992. Effects of C6 = C10 alkenals and alkanals on eliciting a defence response in the developing cotton boll. *Phytochemistry* 31: 2305–2308.
- Zhou S, Lou Y-R, Tzin V, Jander G. 2015. Alteration of plant primary metabolism in response to insect herbivory. *Plant Physiology* 169: 1488–1498.
- Zhu-Salzman K, Salzman RA, Ahn J-E, Koiwa H. 2004. Transcriptional regulation of Sorghum defense determinants against a phloem-feeding aphid. *Plant Physiology* 134: 420–431.
- Ziegler J, Keinänen M, Baldwin IT. 2001. Herbivore-induced allene oxide synthase transcripts and jasmonic acid in *Nicotiana attenuata*. *Phytochemistry* 58: 729–738.

Supporting Information

Additional Supporting Information may be found online in the Supporting Information tab for this article:

Fig. S1 Funnel plots for total GLV production after treatment by a fungus, insect or wounding.

Fig. S2 Treatment effects of studies which use wounding + oral secretions or real herbivory.

Table S1 Treatment effect and one-sample, two-sided *t*-test for the different treatments

Table S2 Two-way ANOVA on the treatment effect with treatment (fungus, insect, wounding) and taxa (monocots, eudicots) as fixed factors

Methods S1 Overview of the methodology used to search online databases for publications on GLV production.

Methods S2 Overview of the methodology used to perform the meta-analysis and investigate potential publication bias and robustness of data.

Notes S1 Overview of the publications that were employed to compile the dataset used for the meta-analysis.

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